

Stable Rotating Waves in Rings of a Modified Ventricular Myocyte Model Near a Hopf Point: Computer-Assisted Proofs in Fourier Space

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Abstract

We study an 18-dimensional modification, due to Erhardt, of the ten Tusscher–Panfilov 2006 endocardial ventricular cell model, with reduced repolarization reserve and the slow delayed rectifier conductance $G_{Ks} = 0.0275$ nS/pF, a value about 1.5% below the supercritical Hopf point that Erhardt computed numerically for this model. For the single cell we give two computer-assisted proofs, sharing no library, that a periodic orbit exists and is locally orbitally asymptotically stable: a time-domain proof with CAPD, which encloses the period in $[53.5855190480139, 53.585519630722438]$ ms and bounds the 17 nontrivial Floquet multipliers of its orbit in modulus by 0.998642, and a space-time Fourier proof in Arb ball arithmetic, which encloses the period of its orbit in an interval of width less than 2×10^{-25} ms and bounds its 17 nontrivial multipliers by 0.99785888. An exact rational check shows that a point of the second orbit lies in the ball that the first proof shows the return map to map into itself as a contraction, so the two proofs are about the same orbit. For rings of $N = 8, 16, 32$ and 64 identical cells with voltage-only diffusive coupling of strength $N^2/64000$ per ms we prove, in Fourier space, that a rotating 1-wave $x_j(t) = \phi(\omega t + 2\pi j/N)$ exists, is locally unique, has minimal period enclosed in an interval of width less than 2×10^{-25} ms, is not synchronous, and is locally exponentially orbitally stable with asymptotic phase: the Floquet multiplier 1 is algebraically simple, and the other $18N - 1$ multipliers have modulus less than $e^{-\delta T}$ with $\delta = 5 \times 10^{-6}$ per ms. Existence is proved by a radii-polynomial argument in a weighted ℓ^1 space, with rigorous strip covers, aliasing bounds and a polydisc Cauchy majorant for the non-polynomial ionic currents. Stability is proved through one Hill operator: we show that its spectrum on a half-open strip of height ωN gives every Floquet multiplier of the ring with its algebraic multiplicity, and we exclude spectrum from $\{\operatorname{Re} \mu \geq -\delta\}$ apart from a simple eigenvalue 0 by a Riesz-projection homotopy with a Schur-complement small-gain test and an explicit tail resolvent bound. The coupling of Fourier mode m is an entire function of $\varepsilon = 1/N^2$ whose value at $\varepsilon = 0$ is that of the continuum cable $u_t = f(u) + D u_{xx}$ (voltage only, $D = 1/64000$ per ms, on a ring of unit length). Treating $\varepsilon \in [0, 1/64]$ as an interval parameter, covered by 73 pieces glued by ball inclusion, we also prove that a locally unique rotating 1-wave exists for every $N \geq 8$ and a locally unique traveling wave for the cable, and that they form one family, continuous in ε , so that the ring waves converge to the cable wave as $N \rightarrow \infty$. Their stability is proved only for $N = 8, 16, 32$ and 64 . The identification of these orbits with the Hopf branch is numerical, and nothing is claimed for tissue.

Keywords: cardiac cell model; modified ten Tusscher–Panfilov model; ring of coupled oscillators; rotating wave; Floquet multipliers; Hill operator; computer-assisted proof; ball arithmetic; radii polynomial.

MSC 2020: 37C27, 34C25, 65G20, 92C30, 37M20.

1 Introduction

Detailed ionic models of cardiac cells are systems of a few dozen nonlinear ordinary differential equations whose right-hand sides contain exponentials, logarithms (Nernst potentials) and square roots. Their bifurcations are studied by numerical continuation. Erhardt and Solem analyzed reduced versions of the ten Tusscher–Panfilov 2006 (TP06) model [32] in this way [13, 14], Kügler and coauthors related early afterdepolarizations to period doubling cascades and to folded nodes [27, 28], and Erhardt [12] studied an 18-dimensional modification of the TP06 endocardial cell with reduced repolarization reserve, continued in the slow delayed rectifier conductance G_{Ks} . For that model he reports a supercritical Hopf bifurcation at $G_{Ks} = 0.027907858929580$ nS/pF with stable bifurcating cycles [12, Table 2].

This paper gives computer-assisted proofs for that model at $G_{Ks} = 0.0275$, about 1.46% below the Hopf value, for the single cell and for rings of N identical cells coupled diffusively through the membrane potential. The ring is the simplest spatial object built from these cells: a periodic chain $x_0, \dots, x_{N-1}$ with

$$\dot{x}_j = f(x_j) + c E(x_{j-1} - 2x_j + x_{j+1}), \quad j \in \mathbb{Z}_N, \quad c = \frac{N^2}{64000} \text{ per ms}, \quad (1)$$

where f is the cell’s vector field and E the projection on the voltage. The coupling $c = N^2 D$ with $D = 1/64000$ is that of a ring of unit length discretized into N cells. The ring is equivariant under the cyclic shift, and a rotating wave (a discrete traveling wave) is a solution $x_j(t) = \phi(\omega t + 2\pi j/N)$ with ϕ 2π -periodic. That such waves exist near a Hopf point of a ring of identical cells is the generic expectation of $\mathbb{Z}_N$ -equivariant Hopf theory [18], [19, Ch. XVIII, Sec. 4]; we make no claim that their existence is surprising. What we add is a proof, at an explicit parameter value and explicit coupling strengths, of existence, local uniqueness, the minimal period and local exponential orbital stability, for the full 18-dimensional nonlinear cells.

Results. Theorem A concerns the single cell. It has two parts, proved independently: one in the time domain with CAPD [23], by a contraction argument for the Poincaré map, and one in Fourier space with Arb [21], by the method of this paper with $N = 1$. The two proofs share no library, and an exact rational check (Lemma 6.1) shows that they enclose the same orbit. We call the existence proof in Fourier space Stage E (Section 4) and the stability proof Stage S (Section 5), as the programs and their records do. Theorem B concerns the rings with $N = 8, 16, 32, 64$. Its existence part (Section 4) is a radii-polynomial (Newton–Kantorovich) argument for the Fourier coefficients of the profile ϕ in a weighted ℓ^1 space. In Fourier space the ring enters only through a scalar damping $d_m = 4c \sin^2(\pi m/N)$ on the voltage of mode m , so the cost does not grow with the dimension $18N$ of the ring: each existence proof takes about two minutes. Its stability part (Section 5) rests on a lemma (Theorem 5.3) that the Floquet multipliers of the rotating wave, with algebraic multiplicities, are the numbers $e^{\mu T}$ for μ in the spectrum of a single Hill operator H_0 in a half-open strip of height ωN . Spectrum of H_0 is then excluded from the half plane $\{\operatorname{Re} \mu \geq -\delta\}$, except a simple eigenvalue at 0 and its translates by $i\omega N\mathbb{Z}$, by counting the rank of a Riesz projection along a homotopy (Theorem 5.15). The stability proofs take between half a minute and eight minutes. Theorem C extends existence and local uniqueness to every ring size $N \geq 8$ and to the continuum cable. The damping d_m is an entire function of $\varepsilon = 1/N^2$, and its value at $\varepsilon = 0$ is the damping of the cable, so the profile equations of all the rings and of the cable belong to one family (Section 2.3). We prove existence for every $\varepsilon \in [0, 1/64]$ by making the radii-polynomial argument uniform over 73 parameter pieces, in the manner of rigorous parameter continuation [33], and glue the pieces by ball inclusion (Section 4.8). Stability is proved only where Stage S was run, for $N = 8, 16, 32, 64$.

Related work. Space-time Fourier existence proofs for periodic orbits by the radii-polynomial approach are due to Gameiro and Lessard [17], who proved periodic orbits of the Kuramoto–Sivashinsky equation and enclosed individual Floquet exponents by a second contraction on the invariance equation [17, Sec. 7]. That proves instability; they state that their approach could only prove that some solutions are unstable, and that extending it to prove stability “is the subject of current research” [17, Sec. 1]. The exclusion and counting step of Section 5 is how we supply the missing part for our Hill operator. Other routes to the stability of a periodic orbit exist: Castelli and Lessard [5] computed the Floquet normal form $\Phi(t) = Q(t)e^{Rt}$ of periodic orbits of ODEs by rigorous numerics, which gives all Floquet exponents; a time-domain contraction for the Poincaré map, as in Section 6, is another; and Church, Dai, Hénot, Lappicy and Vassena [6] proved global families of stable periodic orbits in a model of competing predators by computer; their abstract does not say how stability is certified. Figueras, Gameiro, Lessard and de la Llave [16] gave a framework for the numerical computation and a posteriori verification of invariant objects of evolution equations. Church and Lessard [7] verified Hopf bifurcations in functional differential equations of mixed type, the class to which the profile equation of a lattice traveling wave belongs, Arioli and Koch [2] constructed traveling waves of the FPU chain by a computer-assisted proof, and Kuehn and Queirolo [26] proved Hopf bifurcation points and periodic orbits of recurrent neural networks by computer. Bayer and Leine [4] proved an explicit a priori bound on the truncation error of the Koopman–Hill approximation of the monodromy matrix of a linear time-periodic system, for coefficients with $\|J_k\| \leq ae^{-b|k|}$ and $b > \ln 2$ [4, Theorem 4], and inferred multiplier locations from pseudospectra in floating point; our tail bound is a posteriori and acts on the resolvent of the Hill operator, which lets us count eigenvalues in ball arithmetic. Johnson and Zumbrun [22] proved convergence of Hill’s method for nonselfadjoint spatially periodic operators; we cite it as background on Hill truncation only, since it gives no explicit error bounds and does not cover the operator $d/dt - A(t)$ of a time-periodic Floquet problem.

The reduction of a ring’s Floquet problem to Fourier blocks combines three classical ingredients: the $\mathbb{Z}_N$ -isotypic decomposition of rings of identical cells [18], [19, Ch. XVIII, Sec. 4]; the spatio-temporal symmetry of a discrete wave, which makes the monodromy an N -th power of a twisted map, as in Rucklidge and Silber [31] and de Wolff [10]; and the discrete Laplacian eigenvalues $4\sin^2(\pi m/N)$, familiar from the master stability function of Pecora and Carroll [30]. We state the resulting identity as a lemma and prove it for completeness. Paullet and Ermentrout [29] studied stable rotating waves in two-dimensional discrete active media analytically, and Ermentrout [15] gave conditions for the stability of phase-locked solutions of weakly coupled oscillator arrays; Ashwin and Swift [3] and Hoppensteadt and Izhikevich [20] treat weakly coupled identical oscillators and phase reduction. For piecewise-linear Chua–Yang rings, Di Marco, Forti, Garay, Koller and Pancioni [11] obtained, according to their abstract, the Floquet multipliers of an unstable, metastable rotating wave. Reentry on rings of excitable (not self-oscillating) cardiac tissue has a long numerical and analytical history, for example Courtemanche, Glass and Keener [8] and ten Tusscher and Panfilov [32]; the waves of this paper are of a different kind, phase waves on rings of self-oscillating cells. For the FitzHugh–Nagumo traveling-wave ODE, Czechowski and Zgliczyński [9] proved periodic orbits by computer, without stability, and, according to them, Arioli and Koch [1] proved the existence and stability of the FitzHugh–Nagumo traveling pulse by computer. Kapela and Zgliczyński [24] proved choreographies of the N -body problem by computer; a choreography has the same cyclic-shift symmetry $x_{j+1}(t) = x_j(t + T/N)$ as a rotating wave, in a Hamiltonian system without diffusive coupling.

Novelty. The searches we made (Section 9) found no earlier computer-assisted proof of a periodic orbit of a detailed ionic cardiac cell model, and no earlier computer-assisted proof of a rotating wave in a ring of diffusively coupled cells. We make no claim beyond that sentence; the nearest computer-assisted precedents for the symmetry and for stability are cited above. The oscillation of the single cell is not new: Erhardt’s continuation [12] predicts it.

Labels. Theorems A, B and C are computer-assisted: every inequality they rest on is decided in interval or ball arithmetic by the programs of Section 8, and every step from those inequalities to the statements is proved in this paper. The lemmas are proved here; so are the standard facts about operators with compact resolvent that Section 5 uses, in Appendix A, from elementary facts only. Section 7 collects numerical observations, labelled as such, among them a floating-point comparison with the weak-coupling phase reduction (Remark 7.1); none of them is used in a proof or stated as a theorem. The checks made of this work are listed in Section 9.

2 The model and the ring

2.1 The cell

The vector field $f : U \subset \mathbb{R}^{18} \rightarrow \mathbb{R}^{18}$ is the function `fun_eval` of the file `bifurcation analysis/TP06_18d_endo_bif.m` in Erhardt’s MIT-licensed repository `andreerhardt/cardiac-dynamics-of-a-human-ventricular-tissue-model-with-focus-on-early-afterdepolarizations`, commit `dc78f86fd218418e029ec43d945bcd0fc54b9f1e`, file SHA-256 `a50f6c08b4360dd257cce389a39ae72fda51e3642641bf5b8e5fced6c2225670`, which accompanies [12]. The state is $(V, X_{r1}, X_{r2}, X_s, m, h, j, d, f, f_2, f_{Ca_{ss}}, s, r, R', Ca_i, Ca_{sr}, Ca_{ss}, Na_i)$, in mV and mM, and time is in ms. It is Erhardt’s 18-state modification of the TP06 endocardial cell [32], and we call it that rather than “the TP06 cell”, because it differs from the published TP06 cell in four ways:

1. the intracellular potassium concentration K_i is a parameter held at 138.3 mM, so 19 states become 18;
2. the Heaviside switch at $V = -40$ mV in the rates of the gates h and j is replaced by $u(V) = 1/(1 + e^{-5(V+40)})$, which makes f real analytic;
3. Erhardt’s reduced-repolarization parameters: $G_{Kr} = 0.0153$ (0.1 times the endocardial value), $G_{CaL} = 0.000199$ (5 times), with G_{Ks} the continuation parameter, here $G_{Ks} = 0.0275$; also $G_{Na} = 14.838$, $G_{K1} = 5.405$ and no stimulus;
4. Erhardt’s capacitance convention: the parameter $C_m = 1$ divides dV/dt and also multiplies the fluxes of Ca_i , Ca_{ss} and Na_i , where the original authors’ convention (and the CellML version) uses the numerical value 0.185 for the cell capacitance (185 pF); so every concentration flux is $1/0.185 \approx 5.405$ times its value in that convention. We state only this ratio and make no claim about the physical units of Erhardt’s C_m .

The domain U is the open set on which every logarithm argument is positive, every square-root argument is positive and every divisor is nonzero. In the proofs every decimal constant of the source enters as an exact rational (Section 8), so the theorems are about the model as written, not about its floating-point rounding. All computations use the scaled variables $z = \Sigma^{-1}x$, $\Sigma = \text{diag}(2^{e_1}, \dots, 2^{e_{18}})$ with the scale exponents $(-2, 0, -5, -1, 0, -28, -28, 0, -4, -2, -1, -8, -5, -1, -10, 2, -3, 3)$ (`code/model/scales.txt`); a constant diagonal change of variables commutes with E and changes no

eigenvalue or multiplier, and we keep writing f for the field in scaled variables where no confusion arises.

2.2 The ring and its symmetry

For $N \geq 1$ the ring is (1) on $(\mathbb{R}^{18})^N$, with $E = e_V e_V^\top$ the projection on the voltage component; for $N = 1$ the coupling vanishes and (1) is the cell. Write F for its vector field and Q for the shift $(Qx)_j = x_{j+1}$. Then $F(Qx) = QF(x)$ and $Q^N = I$.

Let $\omega > 0$, $T = 2\pi/\omega$, $\tau = T/N$, and let $\phi : \mathbb{R} \rightarrow \mathbb{R}^{18}$ be 2π -periodic, real analytic and nonconstant, with $\phi(\theta) = \sum_{m \in \mathbb{Z}} a_m e^{im\theta}$. Write $\theta_j(t) = \omega t + 2\pi j/N$ and $(\Delta_N p)(\theta) = p(\theta - 2\pi/N) - 2p(\theta) + p(\theta + 2\pi/N)$. If ϕ solves the profile equation

$$\omega \phi'(\theta) = f(\phi(\theta)) + c E (\Delta_N \phi)(\theta) \quad (\theta \in \mathbb{R}), \quad (2)$$

then $x_j(t) = \phi(\theta_j(t))$ solves (1), because $x_{j\pm 1}(t) = \phi(\theta_j(t) \pm 2\pi/N)$, and indices modulo N are consistent because ϕ is 2π -periodic. Moreover $x(t + \tau) = Qx(t)$, since $\theta_j(t + \tau) = \theta_{j+1}(t)$. We call x a rotating 1-wave. On mode m the operator $c E \Delta_N$ acts as $-d_m E$ with

$$d_m = 4c \sin^2(\pi m/N), \quad 0 \leq d_m \leq 4c, \quad d_{-m} = d_m = d_{m+N} = d_{N-m}, \quad (3)$$

so in Fourier space (2) reads $i\omega m a_m = [f \circ \phi]_m - d_m E a_m$ for every $m \in \mathbb{Z}$. The ring is felt only through the scalars d_m .

2.3 One family for all rings and the cable

Let $D = 1/64000$ per ms, so that $c = N^2 D$. For $\varepsilon \geq 0$ and $m \in \mathbb{Z}$ let

$$d_m(\varepsilon) = 4\pi^2 D m^2 S(\pi^2 m^2 \varepsilon)^2, \quad S(w) = \sum_{k \geq 0} \frac{(-w)^k}{(2k+1)!}. \quad (4)$$

S is entire with real coefficients, and $S(w) = \sin \sqrt{w}/\sqrt{w}$ for $w > 0$. So each d_m is a real analytic function on $\mathbb{R}$, $d_m(\varepsilon) \geq 0$ for $\varepsilon \geq 0$, $d_{-m} = d_m$, $d_0 = 0$, $d_m(0) = 4\pi^2 D m^2$, and $d_m(1/N^2) = 4N^2 D \sin^2(\pi m/N)$, which is (3). For $\varepsilon = h^2 > 0$, $d_m(\varepsilon) = 4D \sin^2(\pi m h)/h^2$, and $-d_m(\varepsilon)E$ is the action on mode m of

$$(\mathcal{L}_\varepsilon p)(\theta) = \frac{D}{h^2} E (p(\theta - 2\pi h) - 2p(\theta) + p(\theta + 2\pi h)) \quad (\varepsilon = h^2 > 0), \quad \mathcal{L}_0 p = 4\pi^2 D E p'', \quad (5)$$

where $\mathcal{L}_0$ is understood for C^2 functions p . The profile equation of the family is

$$\omega \phi'(\theta) = f(\phi(\theta)) + (\mathcal{L}_\varepsilon \phi)(\theta) \quad (\theta \in \mathbb{R}). \quad (6)$$

For $\varepsilon = 1/N^2$, $\mathcal{L}_\varepsilon = c E \Delta_N$ and (6) is (2). For $\varepsilon = 0$, if ϕ is C^2 , 2π -periodic and solves (6), then $u(x, t) = \phi(\omega t + 2\pi x)$ solves the cable equation

$$u_t = f(u) + D E u_{xx}, \quad x \in \mathbb{R}/\mathbb{Z}, \quad (7)$$

because $u_t = \omega \phi'$ and $u_{xx} = 4\pi^2 \phi''$; the ring (1) is the discretization of (7) by N cells of length $1/N$. For other $\varepsilon > 0$, (6) is the profile equation of a lattice of spacing $\sqrt{\varepsilon}$, which closes into a ring only when $1/\sqrt{\varepsilon}$ is an integer; these values of ε serve only to connect the rings with each other and with the cable.

3 Results

Let s be the double-precision number nearest to 0.2, and call the section $\{V = s\}$ (in mV).

Theorem A (the cell; computer-assisted). *Let $N = 1$ and $G_{Ks} = 0.0275$.*

- (i) (CAPD, time domain.) *The first-return map g to the section $\{V = s\}$, crossed upward, has a unique fixed point x_c in the explicit ball B of Section 6. The periodic orbit through x_c has period in $[53.5855190480139, 53.585519630722438]$ ms (exact bounds `0x1.acaf249c533cfp+5` and `0x1.acaf24ea88f37p+5`), every eigenvalue of $Dg(x_c)$, that is every one of the 17 nontrivial Floquet multipliers, has modulus at most $0.99864150686405151 < 0.998642$, and the orbit is locally orbitally asymptotically stable.*
- (ii) (Arb, Fourier space.) *There are $\omega_* > 0$ and a real analytic 2π -periodic ϕ_* with $\phi_{*,V}(0) = s$ such that $x(t) = \phi_*(\omega_* t)$ is a periodic orbit of minimal period $T = 2\pi/\omega_*$ with*

$$T \in [53.58551933936116920991898, 53.58551933936116920991899] \text{ ms}$$

(23-decimal outward roundings of binary ends 1.56×10^{-25} ms apart); the pair $(\omega_, (a_m))$ is locally unique in the sense of Theorem B(a) with $N = 1$. The Floquet multiplier 1 is algebraically simple and the other 17 multipliers have modulus less than $e^{-\delta T} \leq 0.9978588747123748493822038$, where $\delta = 23058430092137/2^{59} \approx 4.0000000000000104916 \times 10^{-5}$ per ms; the orbit is locally exponentially orbitally stable with asymptotic phase.*

- (iii) (Link.) *The orbits of (i) and (ii) are the same: $\Sigma^{-1}\phi_*(0) \in B$, so x_c lies on the orbit of (ii) (Lemma 6.1). The period interval of (ii) lies inside that of (i).*

Each of (i) and (ii) assumes that its program evaluates Erhardt's function with exact decimals (the CAPD field in (i), the Arb field in (ii)); (iii) uses both assumptions together, so a fault in either translation would also break the link. The two programs were compared at 104 points by the tests (Section 8), not proved equal (Section 9).

Theorem B (rings; computer-assisted). *Let $G_{Ks} = 0.0275$, $N \in \{8, 16, 32, 64\}$ and $c = N^2/64000$ per ms.*

- (a) (Existence and local uniqueness.) *There are $\omega_* > 0$ and a real 2π -periodic ϕ_* , analytic on the strip $|\operatorname{Im} \theta| < 1/4$, with $\phi_{*,V}(0) = s$, such that $x_j(t) = \phi_*(\omega_* t + 2\pi j/N)$ solves (1). The pair $(\omega_*, (a_m)_{m \in \mathbb{Z}})$ formed by ω_* and the Fourier coefficients of $\Sigma^{-1}\phi_*$ (the profile in the scaled variables of Section 2.1) lies within r_{ex} of an explicit center in the norm $\max(|\omega|, \max_k \sum_m |a_{k,m}|e^{|m|/4})$, and it is the only zero of the map (8) below in the ball of radius r_{un} about that center, where r_{un} is the largest double not above 10^{-12} ($r_{\text{un}} \approx 9.999999999999997989 \times 10^{-13}$); r_{ex} is given in Table 1.*
- (b) (Period, minimal period, wave number.) *$T = 2\pi/\omega_*$ lies in a certified interval of width less than 2×10^{-25} ms (between 1.49×10^{-25} and 1.51×10^{-25} for the four values of N), which Table 1 prints rounded outward to 23 decimals. Each cell has minimal period T , the solution is not synchronous (it is not the case that $x_j = x_0$ for all j), and it is a rotating 1-wave: $x_j(t) = x_0(t + jT/N)$, and no relation $x_j(t) = x_0(t + jkT/N)$ for all j holds with $k \not\equiv 1 \pmod{N}$.*

(c) (Stability.) Let $\delta = 5764607523035/2^{60} \approx 5.0000000000006636358 \times 10^{-6}$ per ms. The monodromy matrix of the linearization along the wave over the period T has the eigenvalue 1 algebraically simple, and its other $18N - 1$ eigenvalues (the nontrivial Floquet multipliers, with algebraic multiplicity) have modulus less than $e^{-\delta T}$, which is at most the bound of Table 2. The shift-reduced map $Q^{-1}\varphi_\tau$, $\tau = T/N$, has derivative at $x(0)$ with eigenvalue 1 algebraically simple and all other eigenvalues of modulus less than $e^{-\delta\tau}$. The orbit is locally exponentially orbitally stable with asymptotic phase: for every $\delta' < \delta$ there are $\epsilon_0 > 0$ and C such that every solution starting within distance ϵ_0 of the orbit O satisfies, for some $\sigma \in \mathbb{R}$ and all $t \geq 0$, $|\varphi_t(x^0) - x(t + \sigma)| \leq Ce^{-\delta't} \text{dist}(x^0, O)$.

Here φ_t is the flow of (1). The constants ϵ_0 and C are not computed.

N	lower end of the enclosure of T (ms)	r_{ex}	Z_1
1	53.58551933936116920991898	1.7073×10^{-28}	0.20366
8	53.58797097681944967415082	1.6440×10^{-28}	0.20339
16	53.58806910359016923593730	1.6415×10^{-28}	0.20338
32	53.58809413031803250655154	1.6409×10^{-28}	0.20337
64	53.58810041831757754104213	1.6408×10^{-28}	0.20337

Table 1: Existence (Section 4), from the existence records in `data/`. The printed ends are the exact binary ends stored in the record, about 1.5×10^{-25} ms apart, rounded outward to 23 decimals; the upper end is one unit larger in the last printed digit. r_{ex} (rounded up here) and Z_1 (rounded) are the radius at which the radii polynomial is negative and the bound of Lemma 4.5; the uniqueness radius is $r_{\text{un}} \approx 10^{-12}$ for every N , set by the validity radius r_* of the bound Z_2 .

N	bound on $e^{-\delta T}$ (full period)	bound on $e^{-\delta\tau}$	K_e	θ_T	worst (SC) ratio
1	0.9978588747123748493822038	(same)	16	0.2948	0.104
8	0.9997320960377930203747358	0.9999665080790063923613739	20	0.2817	0.090
16	0.9997320955472906096505621	0.9999832538686231493032797	24	0.2519	0.084
32	0.9997320954221904942011761	0.9999916268953467731405854	32	0.2895	0.154
64	0.9997320953907589193958104	0.9999958134384184917812441	48	0.2950	0.169

Table 2: Stability (Section 5), from the stability records in `data/`: the certified upper bounds, rounded upward from Arb, with $\delta = 23058430092137/2^{59}$ for $N = 1$ and $\delta = 5764607523035/2^{60}$ for the rings; the window half-width K_e ; the tail constant $\theta_T < 1$ of Lemma 5.12; and the largest ratio of the left side to the right side of the window inequality (SC) over all window columns, which must be below 1.

Remark 3.1. The CAPD record `data/cell-gks0.0275.json` and the Fourier record `data/fourier-existence-N1.json` store both period enclosures, and the second lies inside the first; Lemma 6.1 shows that the two orbits are the same. An independent computation within the project, by a separate CAPD pipeline that is not part of these proofs, enclosed the periods of the cell in $[53.58551856, 53.58552012]$ ms, of the 8-cell wave in $[53.58795907, 53.58798288]$ ms and of the 16-cell wave in $[53.58805554, 53.58808267]$ ms, with all nontrivial multipliers of modulus below 999/1000 (cell) and 9999/10000 (rings). The periods of Theorem A(ii) and Theorem B for $N = 1, 8, 16$ lie inside these intervals. We use these intervals as a consistency check only.

Theorem C (every ring size and the cable; computer-assisted). Let $G_{Ks} = 0.0275$, and let $d_m(\varepsilon)$, $\mathcal{L}_\varepsilon$ and $D = 1/64000$ per ms be as in Section 2.3.

- (a) *(One continuous family.)* For every $\varepsilon \in [0, 1/64]$ there are $\omega_*(\varepsilon) > 0$ and a real 2π -periodic $\phi_*(\cdot; \varepsilon)$ of minimal period 2π , analytic on the strip $|\operatorname{Im} \theta| < 1/4$, with $\phi_{*,V}(0; \varepsilon) = s$, that solve (6). Let $x_*(\varepsilon)$ be the pair formed by $\omega_*(\varepsilon)$ and the Fourier coefficients of $\Sigma^{-1}\phi_*(\cdot; \varepsilon)$. The map $\varepsilon \mapsto x_*(\varepsilon)$ is continuous from $[0, 1/64]$ into the space X of Section 4.2. The interval $[0, 1/64]$ is covered by 73 pieces, each with an explicit center, and for every piece P that contains ε : $x_*(\varepsilon)$ lies within 9.07×10^{-7} of the center of P in the norm of Theorem B(a); it is the only zero of $F(\cdot; \varepsilon)$ (Section 4.8) within 1.05×10^{-6} of that center in that norm, and within $r_{\text{un}}(P) \geq 4.65 \times 10^{-4}$ of it in a weighted norm stored with the piece; and $T(\varepsilon) = 2\pi/\omega_*(\varepsilon)$ lies in an explicit interval of width less than 1.24×10^{-5} ms.
- (b) *(Every ring size.)* For every integer $N \geq 8$, with $c = N^2/64000$ per ms and $\omega_* = \omega_*(1/N^2)$, $x_j(t) = \phi_*(\omega_*t + 2\pi j/N; 1/N^2)$ solves (1). Each cell has minimal period $T_N = T(1/N^2)$, the solution is not synchronous, and it is a rotating 1-wave in the sense of Theorem B(b). It is locally unique in the sense of (a) with $\varepsilon = 1/N^2$, for which $F(\cdot; \varepsilon)$ is the map (8) of this N . For $N = 8, 16, 32$ and 64 it is the wave of Theorem B, and so it is stable in the sense of Theorem B(c). For every other N its stability is not claimed.
- (c) *(The cable.)* With $\omega_* = \omega_*(0)$, $u(x, t) = \phi_*(\omega_*t + 2\pi x; 0)$, $x \in \mathbb{R}/\mathbb{Z}$, is a real analytic, hence classical, solution of the cable equation (7) on the ring of unit length. It is a traveling wave that goes once around the ring in each period $T_\infty = T(0)$, which is its minimal period in t and lies in $[53.58809529823299300232974, 53.58810763684487454838746]$ ms. It is locally unique in the sense of (a) with $\varepsilon = 0$. Its stability is not claimed.
- (d) *(The limit $N \rightarrow \infty$.)* $x_*(1/N^2) \rightarrow x_*(0)$ in X as $N \rightarrow \infty$. Hence $\phi_*(\cdot; 1/N^2) \rightarrow \phi_*(\cdot; 0)$ uniformly on the closed strip $|\operatorname{Im} \theta| \leq 1/4$, and $T_N \rightarrow T_\infty$.

The pieces, their centers, weights, radii and period enclosures are stored in `data/fourier-existence-alln.json`; Table 3 lists the pieces that contain $1/N^2$ for $N = 8, 16, 32, 64$, and the cable. For these four N , Theorem B encloses T_N in intervals about 10^{20} times narrower; what Theorem C adds is every other $N \geq 8$, the cable, and the continuity in ε . It says nothing about a rate of convergence in (d). The proof is in Section 4.8.

N	piece P	enclosure of T_N from P (ms)	left side of (11)	$r_{\text{un}}(P)$
8	[63/4096, 1/64]	[53.58796640015, 53.58797756550]	6.90×10^{-7}	4.86×10^{-4}
16	[63/16384, 67/16384]	[53.58806282472, 53.58807434393]	3.57×10^{-7}	4.86×10^{-4}
32	[7/8192, 9/8192]	[53.58808831963, 53.58809994100]	1.31×10^{-11}	4.85×10^{-4}
64	[0, 1/4096]	[53.58809529823, 53.58810763685]	7.25×10^{-7}	4.65×10^{-4}
64	[7/32768, 15/32768]	[53.58809380464, 53.58810545862]	5.40×10^{-7}	6.06×10^{-4}
∞ (cable)	[0, 1/4096]	[53.58809529823, 53.58810763685]		4.65×10^{-4}

Table 3: Theorem C at the ring sizes of Theorem B and for the cable, from `data/fourier-existence-alln.json`. The period enclosures are the record's outward-rounded decimal ends, rounded outward again to 11 decimals; each contains the enclosure of Table 1 for the same N . The fourth column (rounded up) is the left side of the inclusion (11) of the existence ball of Theorem B(a) in the uniqueness ball of P , which must not exceed $r_{\text{un}}(P)$ (rounded down); $1/4096$ lies in two pieces, and both inclusions hold.

4 Existence in Fourier space

4.1 Rigorous Fourier coefficients of a composition

Let $p(\theta) = \sum_{|m| \leq K} a_m e^{im\theta}$ with $a_m \in \mathbb{C}^n$ given as complex balls; every statement below holds for every trigonometric polynomial with coefficients in the balls. Let $f : U \rightarrow \mathbb{C}^q$ be holomorphic on an open $U \subset \mathbb{C}^n$, given by an inclusion function F : a program that maps a list of complex balls X to balls Y or raises an error, such that whenever all of Y is finite, $X \subset U$ and $f(z) \in Y$ for every $z \in X$. A program written with $+$, $-$, $\times$, integer powers, $\exp$, division, and logarithms and square roots that raise unless their argument ball lies in $\{\operatorname{Re} w > 0\}$ is such an inclusion function, with U the set on which every divisor is nonzero and every logarithm and square-root argument has positive real part, by induction over the expression. (Arb returns a non-finite ball for a quotient whose divisor ball contains 0; Arb 's own complex logarithm does not signal a ball that crosses the branch cut, which is why the guards are needed.) Let $g = f \circ p$, $c_k = \frac{1}{2\pi} \int_0^{2\pi} g(\theta) e^{-ik\theta} d\theta$, and $\Sigma_\rho = \{\theta \in \mathbb{C} : |\operatorname{Im} \theta| \leq \rho\}$.

Lemma 4.1 (strip cover). *Cover the rectangle $R = [0, 2\pi] \times [-\rho, \rho]$ by finitely many closed rectangles B with exact rational corners. For each B let Θ_B be a complex ball containing B , $P_B = \sum_m a_m \exp(im\Theta_B)$ evaluated in ball arithmetic, and $Y_B = F(P_B)$. If every Y_B is finite, then $p(\Sigma_\rho) \subset U$, g is holomorphic on an open neighborhood of Σ_ρ , and $\sup_{\Sigma_\rho} |g_i| \leq S_i := \max_B \overline{|Y_{B,i}|}$, where $\overline{|\cdot|}$ is an upper bound of the modulus.*

Proof. Every $\theta \in R$ lies in some B , so $p(\theta) \in P_B \subset U$ by the inclusion property. Since p is 2π -periodic, $p(\Sigma_\rho) = p(R) \subset U$; as U is open and p continuous, $p^{-1}(U)$ is an open set containing Σ_ρ , on which g is holomorphic. Finally $|g_i(\theta)| \leq \overline{|Y_{B,i}|} \leq S_i$. $\square$

The whole rectangle is covered, not only its edges: a pole of g strictly inside the strip can coexist with a finite supremum on the edges, and then the next lemma fails. The programs refine the cover adaptively and optionally use a centered (mean-value) form on each box; both only affect how tight S is.

Lemma 4.2 (Cauchy estimate). *If g is holomorphic on an open set containing Σ_ρ , 2π -periodic, and $|g| \leq S$ on Σ_ρ , then $|c_m| \leq S e^{-\rho|m|}$ for every $m \in \mathbb{Z}$.*

Proof. For $m > 0$ integrate $g(\theta) e^{-im\theta}$ over the boundary of the rectangle with corners $0, 2\pi, 2\pi - i\rho, -i\rho$. The integral vanishes by Cauchy's theorem and the vertical sides cancel by periodicity, so $c_m = \frac{1}{2\pi} \int_0^{2\pi} g(x - i\rho) e^{-im(x-i\rho)} dx$ and $|e^{-im(x-i\rho)}| = e^{-m\rho}$. For $m < 0$ shift to $\operatorname{Im} \theta = \rho$; $|c_0| \leq S$ directly. $\square$

Lemma 4.3 (aliasing). *Under the hypotheses of Lemma 4.2, with $\hat{c}_k = \frac{1}{M} \sum_{j=0}^{M-1} g(2\pi j/M) e^{-2\pi ijk/M}$ and $|k| < M$,*

$$|\hat{c}_k - c_k| \leq E_k := S \frac{e^{-\rho(M-k)} + e^{-\rho(M+k)}}{1 - e^{-\rho M}}.$$

Proof. By Lemma 4.2 $\sum_m |c_m| < \infty$, so the Fourier series converges absolutely and uniformly to g . Substituting and using $\frac{1}{M} \sum_j e^{2\pi i j(m-k)/M} = 1$ if $m \equiv k \pmod{M}$ and 0 otherwise gives $\hat{c}_k = \sum_{l \in \mathbb{Z}} c_{k+lM}$. For $l \geq 1$, $k+lM > 0$ and $\sum_{l \geq 1} |c_{k+lM}| \leq S e^{-\rho(M+k)} / (1 - e^{-\rho M})$; for $l = -j \leq -1$, $|k - jM| = jM - k$ and the sum is at most $S e^{-\rho(M-k)} / (1 - e^{-\rho M})$. $\square$

Proposition 4.4 (enclosure of the coefficients). *Let the strip cover succeed with bounds S_i . Evaluate the nodes $e^{im\theta_j}$, $\theta_j = 2\pi j/M$, as balls (e^{iq} for exact rationals q), $P_j = \sum_m a_m e^{im\theta_j}$ and $G_j = F(P_j)$, and let $C_k = \frac{1}{M} \sum_j G_j e^{-ik\theta_j} + E_k([-1, 1] + i[-1, 1])$. Then for every p with coefficients in the balls, every component i and every $|k| \leq K' < M$, $c_{i,k}[f \circ p] \in C_{i,k}$, and $|c_{i,m}| \leq S_i e^{-\rho|m|}$ for every m .*

Proof. The ball sum contains $\hat{c}_k$, and a complex number w with $|w| \leq E_k$ lies in the square $E_k([-1, 1] + i[-1, 1])$. Lemmas 4.1 to 4.3. $\square$

For $\nu > 0$ with $q = \nu e^{-\rho} < 1$ these bounds give the weighted tail $\sum_{|m| > K'} |c_m| \nu^{|m|} \leq 2Sq^{K'+1}/(1-q)$.

4.2 The zero-finding problem

Fix $K = 32$ and a center $\bar{x} = (\bar{\omega}, \bar{a})$ of exact dyadic numbers, with $\bar{a}_m = 0$ for $|m| > K$, $\bar{a}_{-m} = \overline{\bar{a}_m}$ and $\bar{\omega}$ real; it is computed by an untrusted Newton iteration (`code/fourier/centre.py`) and stored in `code/fourier/data/centre_N*_K32.json`. Let $\nu = e^{1/4}$, let ℓ_ν^1 be the sequences $(b_m)_{m \in \mathbb{Z}}$ with $\|b\|_\nu = \sum_m |b_m| \nu^{|m|} < \infty$, a Banach algebra under convolution, and let $X = \mathbb{C} \times (\ell_\nu^1)^{18}$ with $\|x\| = \max(|\omega|, \max_k \|a_k\|_\nu)$ (the programs allow weights; all runs of Stage E used weight 1, and Section 4.8 uses weights). Writing $\phi_a(\theta) = \sum_m a_m e^{im\theta}$, $g(a) = f \circ \phi_a$ and $g_m(a)$ for its Fourier coefficients, define F , on the open set of $x \in X$ for which ϕ_a maps the closed strip $|\operatorname{Im} \theta| \leq 1/4$ into the domain U (it contains the ball $B_{r_*}(\bar{x})$ used below, Section 4.6), with values in $X' := \mathbb{C} \times (\ell_\nu^1)^{18}$, by

$$F_{\text{ph}}(x) = \sum_m a_{m,V} - s/\sigma_V, \quad F_m(x) = i\omega m a_m - g_m(a) + d_m E a_m \quad (m \in \mathbb{Z}), \quad (8)$$

where $\sigma_V = 2^{-2}$ is the scale of V , so $F_{\text{ph}} = 0$ says $\phi_{a,V}(0) = s$ in physical units. Here ℓ_ν^1 is the space of sequences with $\|b\|' = \sum_m |b_m| \nu^{|m|}/(1+|m|) < \infty$, and X' has the norm $\max(|\cdot|, \max_k \|\cdot\|')$; F maps X into X' because $(i\omega m a_m)_m \in \ell_\nu^1$ when $a \in \ell_\nu^1$, and $g(a)$, $d_m E a_m \in \ell_\nu^1 \subset \ell_\nu^1$. A zero of F with $\omega > 0$ and ϕ_a real and C^1 solves (2), and the phase condition removes the time-shift invariance.

4.3 The radii-polynomial theorem

Let $A : X' \rightarrow X$ be an injective bounded linear operator, defined below, and $\Psi(x) = x - AF(x)$, a map from X to X . Its fixed points are exactly the zeros of F .

Lemma 4.5 (radii polynomial). *Let $r_* > 0$ be such that Ψ is defined and C^1 on the closed ball $B_{r_*}(\bar{x})$, and suppose that*

$$\|AF(\bar{x})\| \leq Y_0, \quad \|I - ADF(\bar{x})\| \leq Z_1, \quad \|A(DF(x) - DF(\bar{x}))\| \leq Z_2 \|x - \bar{x}\| \quad (x \in B_{r_*}(\bar{x})).$$

If $0 < r \leq r_$, $p(r) := Y_0 + (Z_1 - 1)r + Z_2 r^2/2 < 0$ and $Z_1 + Z_2 r < 1$, then Ψ maps $B_r(\bar{x})$ into itself and is a contraction there, so F has exactly one zero in $B_r(\bar{x})$.*

Proof. $\|D\Psi(x)\| \leq Z_1 + Z_2 \|x - \bar{x}\|$. For $x \in B_r(\bar{x})$, $\|\Psi(x) - \bar{x}\| \leq \|\Psi(\bar{x}) - \bar{x}\| + \|\int_0^1 D\Psi(\bar{x} + t(x - \bar{x}))(x - \bar{x}) dt\| \leq Y_0 + Z_1 r + Z_2 r^2/2 < r$. The mean value inequality on the convex ball gives the contraction constant $Z_1 + Z_2 r < 1$, and Banach's fixed point theorem gives one fixed point, which is a zero of F since A is injective. $\square$

The programs check $p(r) < 0$ and $Z_1 + Z_2 r < 1$ in Arb at two exact dyadic radii: the small radius r_{ex} of Table 1, which places the zero, and r_{un} , the largest double not above $r_* = 10^{-12}$, which gives uniqueness (the contraction constant there is about 0.267). The validity radius r_* is chosen by hand, so r_{un} is not an optimal uniqueness radius.

4.4 The operator A and its tail

Let $J(\theta) = Df(\phi_{\bar{a}}(\theta)) = \sum_n J_n e^{in\theta}$. The derivative $DF(\bar{x})y$, $y = (y_\omega, (y_m))$, has phase component $\sum_m y_{m,V}$ and row m equal to $iy_\omega m \bar{a}_m + i\bar{\omega} m y_m - \sum_n J_n y_{m-n} + d_m E y_m$. The operator A is block diagonal with respect to the finite part (the phase row and modes $|m| \leq K$, with the unknown ω) and each tail mode $|m| > K$:

- A_{fin} , an exact complex matrix of size $1 + 18(2K + 1)$, the floating-point inverse of the midpoint of the Galerkin matrix J_{fin} (the restriction of $DF(\bar{x})$ to these rows and columns);
- $A_m = (i\bar{\omega}mI - \hat{J}_0 + d_mE)^{-1}$ for $|m| > K$, where $\hat{J}_0$ is the exact real midpoint of the enclosure of J_0 .

A acts on X' block by block. It is bounded from X' to X because $\sup_{|m|>K} (1 + |m|)|A_m| < \infty$ (Lemma 4.6), and it is injective because every block is: every A_m is an inverse, and A_{fin} is invertible, which follows from $\|I - A_{\text{fin}}J_{\text{fin}}\| \leq Z_1 < 1$, since the compression of $I - ADF(\bar{x})$ to the finite modes is $I - A_{\text{fin}}J_{\text{fin}}$ and the coordinate projection onto the finite modes has norm 1 in X . Since $\hat{J}_0$ and d_m are real, $A_{-m} = \overline{A_m}$, and only $m > 0$ is computed.

Lemma 4.6 (tail of A). *For $K < m \leq m_{\text{max}}$ the matrices A_m are enclosed by Arb inversion of a ball matrix whose only non-exact entries are the narrow balls of d_m (exact for $N = 1$); Arb raises unless every matrix in the ball is certainly invertible, and otherwise returns a ball containing every inverse. Let $G_0 = |\hat{J}_0| + 4cE$ entrywise, $\Upsilon = \bar{\omega}(m_{\text{max}} + 1)$, $G = G_0/\Upsilon$ and $\vartheta = \|G\|_\infty < 1$. Then for every $m > m_{\text{max}}$, entrywise,*

$$|A_m| \leq \frac{1}{\Upsilon} S_G, \quad |mA_m| \leq \frac{1}{\vartheta} S_G, \quad S_G := I + G + G^2 + \frac{\vartheta^3}{1 - \vartheta} \mathbf{1}\mathbf{1}^\top.$$

Proof. With $y = \bar{\omega}m \geq \Upsilon$ and $B_0 = \hat{J}_0 - d_mE$, $|B_0| \leq G_0$ entrywise because $0 \leq d_m \leq 4c$. Then $A_m = (iy)^{-1}(I - B_0/(iy))^{-1} = (iy)^{-1} \sum_k (B_0/(iy))^k$, and entrywise $|A_m| \leq y^{-1} \sum_k (G_0/y)^k \leq \Upsilon^{-1} \sum_k G^k$. Each entry of G^k is at most the corresponding row sum of G^k , which is at most ϑ^k ; summing over $k \geq 3$ gives the last term. Finally $|mA_m| = (y/\bar{\omega})|A_m| \leq \bar{\omega}^{-1} \sum_k (G_0/y)^k$. $\square$

The program increases m_{max} until $\vartheta \leq 1/2$ is certified ($m_{\text{max}} = 391$ at $N = 8$). The suprema $\bar{A}_0 = \sup_{|m|>K} |A_m|$, $\bar{A}_1 = \sup_{|m|>K} |mA_m|$ and $C_n = \sup_{|m|>K} |A_m J'_n|$, with $J'_n = J_n$ for $n \neq 0$ and $J'_0 = J_0 - \hat{J}_0$, are taken entrywise over both ranges; for $|n| \leq 12$, C_n is the entrywise maximum of the explicit products $|A_m J'_n|$ over $K < m \leq m_{\text{max}}$ and of the bound $\Upsilon^{-1} S_G |J'_n| \geq |A_m| |J'_n|$ valid for every $m > m_{\text{max}}$; for $12 < |n| \leq K'$, $C_n = \bar{A}_0 |J'_n|$. Negative m are handled by $A_{-m} J'_n = \overline{A_m J'_n}$.

4.5 The bounds Y_0 and Z_1

The Fourier data come from Proposition 4.4: enclosures of $g_m(\bar{a})$ and J_n for $|m|, |n| \leq K' = 2K + 16 = 80$ from an $M = 192$ node transform, with strip suprema S_g and (entrywise) S_J on $|\text{Im } \theta| \leq \rho = 3/2$, and beyond K' the bounds $|g_{k,m}| \leq S_{g,k} e^{-\rho|m|}$, $|J_{n,jk}| \leq S_{J,jk} e^{-\rho|n|}$. Let $q = \nu e^{-\rho} < 1$. For a linear operator B on X with blocks $B_{cc'}$, $\|B\| \leq \max_c \sum_{c'} \|B_{cc'}\|$ with $\|B_{cc'}\|_{\ell_v^1 \rightarrow \ell_v^1} = \sup_{m'} \sum_m |B(m, m')| \nu^{|m|} / \nu^{|m'|}$ (the column supremum).

Y_0 . The finite part of $AF(\bar{x})$ is $A_{\text{fin}} F_{\text{fin}}(\bar{x})$, computed in Arb. For $|m| > K$, $\bar{a}_m = 0$ and $F_m(\bar{x}) = -g_m$, so $(AF)_m = -A_m g_m$. For $K < |m| \leq K'$ the computed enclosures of A_m are used (all of them are explicit, since $m_{\text{max}} = 391 > K' = 80$), and for $|m| > K'$, entrywise

$|A_m g_m| \leq \bar{A}_0 S_g e^{-\rho|m|}$, so $\sum_{|m|>K'} |(A_m g_m)_c| \nu^{|m|} \leq \sum_k (\bar{A}_0)_{ck} S_{g,k} 2q^{K'+1}/(1-q)$. Then $Y_0 = \max_c (\|(AF)_{c,\text{fin}}\|_\nu + \|(AF)_{c,\text{tail}}\|_\nu)$, the phase/ ω component counting as a component with the single mode 0.

Z_1 . Let $B = I - A D F(\bar{x})$. Finite rows and finite columns give $I - A_{\text{fin}} J_{\text{fin}}$, an Arb matrix product. A tail column (k, m') , $|m'| > K$, meets the finite rows through $-A_{\text{fin}} w$, where w has phase entry 1 if $k = V$ and 0 otherwise (the phase functional sums over all modes) and row (j, m) entry $-J_{m-m',jk}$ (the terms $i\bar{\omega}m$ and d_m vanish off the diagonal $m = m'$); for $K < |m'| \leq K + 16$ the enclosures of J are used, and for $|m'| \geq K + 17$ the bound $|J_{m-m',jk}| \leq S_{J,jk} e^{-\rho(|m'|-\text{sgn}(m')m)}$ makes every term of the weighted column sum divided by $\nu^{|m'|}$ a constant times $\nu^{-|m'|}$ or $(e^{-\rho}/\nu)^{|m'|}$, both decreasing in $|m'|$, so the value at $|m'| = K + 17$ bounds the supremum. For the tail rows, since $A_m(i\bar{\omega}m - \hat{J}_0 + d_m E) = I$,

$$(By)_m = A_m \left[(J_0 - \hat{J}_0)y_m + \sum_{n \neq 0} J_n y_{m-n} \right] = \sum_n A_m J'_n y_{m-n} \quad (|m| > K),$$

(y_ω does not enter because $\bar{a}_m = 0$ there), and $\nu^{|m|} \leq \nu^{|m-n|} \nu^{|n|}$ gives $\|(By)_{c,\text{tail}}\|_\nu \leq \sum_k \mathcal{T}_{ck} \|y_k\|_\nu$ with $\mathcal{T} = \sum_{|n| \leq K'} C_n \nu^{|n|} + \bar{A}_0 S_J 2q^{K'+1}/(1-q)$, for every column. For an output component c and an input component c' let $ff_{cc'}$ be the largest weighted column sum of the finite rows of block (c, c') over the finite columns, and $ft_{cc'}$ the same over the tail columns (bounded as above). A column of B meets the finite rows and the tail rows, so its weighted sum in block (c, c') is at most $\max(ff_{cc'}, ft_{cc'}) + \mathcal{T}_{cc'}$, and

$$Z_1 = \max_c \sum_{c'} (\max(ff_{cc'}, ft_{cc'}) + \mathcal{T}_{cc'}).$$

The ω column has a single finite column and no tail rows ($iy_\omega m \bar{a}_m = 0$ for $|m| > K$), and the ω row of B (the output of the phase column of A_{fin}) has no tail part, since A has no tail in that row, so for it only the finite-row terms enter.

4.6 The bound Z_2 by a polydisc majorant

Let $R_k = 1/1024$ for every component, and let M_k be the strip supremum (Lemma 4.1, on $|\text{Im } \theta| \leq \rho_2 = 1$) of f_k over the family of trigonometric polynomials obtained from $\phi_{\bar{a}}$ by replacing each constant coefficient $\bar{a}_{0,j}$ by the complex box of half-width R_j about it. Since that box contains the disc of radius R_j , $M_k \geq \sup\{|f_k(\phi_{\bar{a}}(\theta) + w)| : |\text{Im } \theta| \leq 1, |w_j| \leq R_j\}$, and the finite cover certifies that f is holomorphic on an open set containing these points.

Lemma 4.7. *Let $q_2 = \nu e^{-\rho_2} < 1$ and $Q_2 = (1 + q_2)/(1 - q_2)$. For $h \in (\ell_\nu^1)^{18}$ with $\|h_j\|_\nu < R_j$, the map $a \mapsto g_k(a)$ is analytic at $\bar{a} + h$ and*

$$\|D^2 g_k(\bar{a} + h)[u, v]\|_\nu \leq \sum_{j,l} \partial_j \partial_l \Phi_k(\|h\|) \|u_j\|_\nu \|v_l\|_\nu, \quad \Phi_k(t) = M_k Q_2 \prod_j (1 - t_j/R_j)^{-1}.$$

Proof. For $|\text{Im } \theta| \leq 1$ the Taylor coefficients $c_\alpha(\theta) = \partial^\alpha f_k(\phi_{\bar{a}}(\theta))/\alpha!$ satisfy $|c_\alpha(\theta)| \leq M_k R^{-\alpha}$ by Cauchy's inequality on the closed polydisc, and they are holomorphic and 2π -periodic in θ ; Lemma 4.2 gives $|[c_\alpha]_m| \leq M_k R^{-\alpha} e^{-|m|}$ and so $\|c_\alpha\|_\nu \leq M_k R^{-\alpha} Q_2$. In the Banach algebra ℓ_ν^1 the series $\sum_\alpha c_\alpha h^\alpha$ (convolution powers) converges absolutely, dominated by $\Phi_k(\|h\|)$. For real θ , $|h_j(\theta)| \leq \|h_j\|_\nu < R_j$, so the Taylor series of f_k at $\phi_{\bar{a}}(\theta)$ converges to $f_k(\phi_{\bar{a}}(\theta) + h(\theta))$; convergence in ℓ_ν^1 is uniform convergence, so the two functions agree and $g_k(\bar{a} + h) = \sum_\alpha c_\alpha h^\alpha$. Differentiating termwise and comparing with the majorant, whose coefficients are nonnegative, gives the bound. $\square$

For $x = \bar{x} + \Delta$ with $\|\Delta\| \leq r$ and $\|y\| \leq 1$,

$$(DF(x) - DF(\bar{x}))y = im(y_\omega \Delta a_m + \Delta \omega y_m) - (Dg(\bar{a} + \Delta a) - Dg(\bar{a}))y_a.$$

The first term $b = y_\omega \Delta a + \Delta \omega y_a$ has $\|b_k\|_\nu \leq 2r$, and $\|A(imb)\|_c \leq \sum_k (N_1 + \bar{A}_1)_{ck} \|b_k\|$, with N_1 the block norms of $A_{\text{fin}} \text{diag}(im)$. The second equals $\int_0^1 D^2 g(\bar{a} + t\Delta a)[\Delta a, y_a] dt$, whose component k has norm at most $rW_k(r)$ with

$$W_k(r) = M_k Q_2 P(r) \left[\left(\sum_j s_j \right)^2 + \sum_j s_j^2 \right], \quad s_j = \frac{1}{R_j - r}, \quad P(r) = \prod_j \frac{R_j}{R_j - r},$$

because $\partial_j P = P/(R_j - t_j)$, $\partial_l \partial_j P = P/((R_j - t_j)(R_l - t_l))$ for $j \neq l$ and $2P/(R_j - t_j)^2$ for $j = l$, and Φ_k is increasing. Hence

$$Z_2 = \max_c \sum_k [2(N_1 + \bar{A}_1)_{ck} + (N_0 + \bar{A}_0)_{ck} W_k(r_*)],$$

with N_0 the block norms of A_{fin} on the coefficient columns; in the ω row only N_0 and N_1 enter. Ψ is defined and C^1 on $B_{r_*}(\bar{x})$ because $r_* < R_j$, g is analytic there, the term $A(i\omega m a_m)$ is a bounded bilinear map ($\sup_{|m| > K} |m| |A_m| < \infty$ by Lemma 4.6), and the other terms are bounded linear.

4.7 The real solution, its period and its wave number

Lemma 4.8. *Let the hypotheses of Lemma 4.5 hold at r_{ex} and r_{un} , and let $x_* = (\omega_*, a_*)$ be the zero. Then ω_* is real and positive, $\phi_* = \phi_{a_*}$ is real-valued, analytic on $|\text{Im } \theta| < 1/4$, and solves (2). If also $|\bar{a}_{1,V}| > r_{\text{ex}}/\nu$, then ϕ_* has minimal period 2π , and for $N \geq 2$ the ring solution $x_j(t) = \phi_*(\omega_* t + 2\pi j/N)$ has minimal period $T = 2\pi/\omega_*$, is not synchronous and is a rotating 1-wave in the sense of Theorem B(b).*

Proof. Let $\kappa(\omega, a) = (\text{conj } \omega, (\text{conj } a_{-m})_m)$, an isometry of X with $\kappa(\bar{x}) = \bar{x}$. The program `code/fourier/tp06_18d_arb.py` that evaluates f is a fixed composition of $+$, $-$, $\times$, $/$, integer powers, 55 exponentials, 4 logarithms and 3 square roots with real constants, and contains no absolute value, real or imaginary part, comparison, branch or conversion. Each of these operations commutes with complex conjugation wherever it is defined: the arithmetic operations algebraically, $\exp$ because its power series has real coefficients, and the principal logarithm and square root on $\{\text{Re } w > 0\}$, the only arguments admitted, because that set is invariant under conjugation and $\text{Log } \bar{w} = \overline{\text{Log } w}$, $\sqrt{\bar{w}} = \overline{\sqrt{w}}$ there. By induction over the expression, if z lies in the domain then so does $\bar{z}$, and $f(\bar{z}) = \overline{f(z)}$. The points used are $\phi_{\bar{a}}(\theta) + h(\theta)$ with θ real and $|h_j(\theta)| < R_j$; this family is closed under conjugation ($\phi_{\bar{a}}(\theta)$ is real) and lies in the certified domain. Hence $g_m(\kappa a) = \overline{g_{-m}(a)}$, and since $d_{-m} = d_m$ and s/σ_V are real, $F(\kappa x) = \overline{F(x)}$ in the corresponding sense; so κ maps zeros of F in $B_{r_{\text{un}}}(\bar{x})$ to zeros in the same ball, and uniqueness gives $\kappa x_* = x_*$: ω_* is real and ϕ_* is real-valued. As $a_* \in (\ell_\nu^1)^{18}$, ϕ_* is analytic on $|\text{Im } \theta| < 1/4$; since $i\omega_* m a_{*,m} = g_m - d_m E a_{*,m}$, the coefficient sequence of ϕ'_* is in ℓ_ν^1 , and (2) holds pointwise. $\omega_* \geq \bar{\omega} - r_{\text{ex}} > 0$ is checked in Arb.

Since $\|a_{*,V} - \bar{a}_V\|_\nu \geq \nu |a_{*,1,V} - \bar{a}_{1,V}|$, $|a_{*,1,V} - \bar{a}_{1,V}| \leq r_{\text{ex}}/\nu$, so $a_{*,1,V} \neq 0$. A function of minimal period $2\pi/k$, $k \geq 2$, has $a_m = 0$ unless $k \mid m$; hence ϕ_* has minimal period 2π and each cell has minimal period T . A synchronous solution would make ϕ_* periodic with period $2\pi/N$. If $x_j(t) = x_0(t + jkT/N)$ for all j and some $k \not\equiv 1 \pmod{N}$, then with $j = 1$, $\phi_*(\theta + 2\pi/N) = \phi_*(\theta + 2\pi k/N)$, so ϕ_* would have the period $2\pi(k-1)/N \not\equiv 0 \pmod{2\pi}$, contradicting minimality. $\square$

Proof of Theorem B(a), (b) and the existence part of Theorem A(ii). For each $N \in \{1, 8, 16, 32, 64\}$ the program `code/fourier/existence.py` computes, in Arb, the strip covers (all leaves finite,

full strip), the enclosures of Proposition 4.4, the tail of Lemma 4.6, Y_0 , Z_1 and Z_2 as above, checks $p(r) < 0$ and $Z_1 + Z_2 r < 1$ at $r = r_{\text{ex}}$ and at $r = r_{\text{un}} \leq r_*$, checks $\bar{\omega} - r_{\text{ex}} > 0$, and $|\bar{a}_{1,V}| > r_{\text{ex}}/\nu$, and writes the interval $2\pi/[\omega_*]$ with outward-rounded decimal ends. The records `data/fourier-existence-N*.json` store every bound; for example at $N = 8$: $Y_0 = 1.3096 \times 10^{-28}$, $Z_1 = 0.20339$, $Z_2 \leq 6.4002 \times 10^{10}$, $|\bar{a}_{1,V}| \geq 0.11524$. Lemmas 4.5 and 4.8 give the statements. $\square$

4.8 Every ring size and the cable

This subsection proves Theorem C. The parameter is ε of Section 2.3, and $G_{Ks} = 0.0275$ is fixed.

Zeros of the family. For $\varepsilon \geq 0$ let $F(x; \varepsilon)$ have the components (8) with d_m replaced by $d_m(\varepsilon)$ of (4): the phase component $F_{\text{ph}}(x)$ and $F_m(x; \varepsilon) = i\omega m a_m - g_m(a) + d_m(\varepsilon) E a_m \in \mathbb{C}^{18}$, $m \in \mathbb{Z}$, defined for every $x = (\omega, a) \in X$ for which ϕ_a maps the closed strip $|\text{Im } \theta| \leq 1/4$ into U . We call x a zero of $F(\cdot; \varepsilon)$ if all these components vanish. For $\varepsilon = 1/N^2$ these are the components of the map (8) of the N -ring, and a zero in the sense of Section 4.2 is a zero in this sense. For $\varepsilon > 0$, $d_m(\varepsilon) \leq 4D/\varepsilon$ is bounded in m ; for $\varepsilon = 0$, $d_m(0)a_m$ grows like $m^2 a_m$, and $F(\cdot; 0)$ need not take values in X' . We therefore work with a fixed-point map that is defined without a codomain for F .

Pieces. A piece is an interval $P = [\varepsilon_{\text{lo}}, \varepsilon_{\text{hi}}]$ with exact dyadic ends, $0 \leq \varepsilon_{\text{lo}} < \varepsilon_{\text{hi}}$, midpoint ε_c and half-width δ_P . On a piece we fix $K = 12$ and: a center $\bar{x} = (\bar{\omega}, \bar{a})$ of exact dyadic numbers, with $\bar{\omega} > 0$, $\bar{a}_m = 0$ for $|m| > K$ and $\bar{a}_{-m} = \overline{\bar{a}_m}$ (an untrusted Newton iterate at ε_c); weights $\eta = (\eta_\omega, \eta_1, \dots, \eta_{18})$, exact dyadic numbers in $(0, 1]$, and the norm $\|x\|_\eta = \max(|\omega|/\eta_\omega, \max_k \|a_k\|_\nu/\eta_k)$ on X ; A_{fin} , the floating-point inverse of the midpoint of the Galerkin matrix of Section 4.4 at $\bar{x}$ with the damping $d_m(\varepsilon_c)$; $\hat{J}_0$ as in Section 4.4; and, for $|m| > K$ and $\varepsilon \in P$, $A_m(\varepsilon) = (i\bar{\omega}mI - \hat{J}_0 + d_m(\varepsilon)E)^{-1}$. Since every weight is at most 1, $\|x\| \leq \|x\|_\eta$, and a weight depends only on the component, so the conjugation κ of the proof of Lemma 4.8 is an isometry of $\|\cdot\|_\eta$. For $\varepsilon \in P$ define $T_\varepsilon(x)$ by

$$T_\varepsilon(x)_{\text{fin}} = x_{\text{fin}} - A_{\text{fin}} F_{\text{fin}}(x; \varepsilon), \quad T_\varepsilon(x)_m = -A_m(\varepsilon) [i(\omega - \bar{\omega})m a_m + \hat{J}_0 a_m - g_m(a)] \quad (|m| > K), \quad (9)$$

where x_{fin} collects ω and the modes $|m| \leq K$, and F_{fin} the phase component and the F_m with $|m| \leq K$.

Lemma 4.9 (tail resolvents for every damping). *Let $\hat{J}_0$ be a real 18×18 matrix, $\bar{\omega} > 0$, $m_{\text{max}} \geq 1$, $\Upsilon = \bar{\omega}(m_{\text{max}} + 1)$, $G = |\hat{J}_0|/\Upsilon$ entrywise, $\vartheta = \|G\|_\infty < 1$, and S_G as in Lemma 4.6. Then for every integer $m > m_{\text{max}}$ and every real $d \geq 0$ the matrix $M = i\bar{\omega}mI - \hat{J}_0 + dE$ is invertible and, entrywise,*

$$|M^{-1}| \leq S_G/\Upsilon, \quad |mM^{-1}| \leq S_G/\bar{\omega}.$$

For $m < -m_{\text{max}}$ the same bounds hold with $|m|$, since M is then the complex conjugate of the matrix for $-m$.

Proof. Let $y = \bar{\omega}m \geq \Upsilon$ and let Λ_y be the diagonal matrix with entry $iy + d$ in the position of V and iy elsewhere, so $M = \Lambda_y - \hat{J}_0$. As d is real, $|iy + d| \geq y$, so $|\Lambda_y^{-1}| \leq I/y$ entrywise and $\|\Lambda_y^{-1} \hat{J}_0\|_\infty \leq \|\hat{J}_0\|_\infty/y \leq \vartheta < 1$. Hence $M^{-1} = \sum_{k \geq 0} (\Lambda_y^{-1} \hat{J}_0)^k \Lambda_y^{-1}$ and, entrywise, $|M^{-1}| \leq y^{-1} \sum_k (|\hat{J}_0|/y)^k \leq \Upsilon^{-1} \sum_k G^k \leq S_G/\Upsilon$, the last step as in the proof of Lemma 4.6. Since $m/y = 1/\bar{\omega}$, $|mM^{-1}| \leq \bar{\omega}^{-1} \sum_k (|\hat{J}_0|/y)^k \leq S_G/\bar{\omega}$. $\square$

Unlike Lemma 4.6, this needs no upper bound on the damping, so it covers $d_m(0) = 4\pi^2 D m^2$. For $K < m \leq m_{\text{max}}$ the program encloses the range of d_m over P in an exact interval $[d_{\text{lo}}, d_{\text{hi}}] \subset [0, \infty)$ (by evaluating (4) on the hull of P), covers it by finitely many balls of width about $\bar{\omega}m/4$, and

encloses A_m on each ball by Arb inversion of the ball matrix (as in Lemma 4.6), so that every $A_m(\varepsilon)$, $\varepsilon \in P$, lies in one of these enclosures. The suprema $\bar{A}_0$, $\bar{A}_1$ and C_n of Section 4.4 are then taken over $|m| > K$ and $\varepsilon \in P$, with the maximum over these enclosures in the explicit range and Lemma 4.9 beyond it.

Lemma 4.10 (the fixed-point map). *Let P be a piece, $\varepsilon \in P$, and B a closed ball about $\bar{x}$ such that, for $x \in B$, ϕ_a maps the closed strip $|\operatorname{Im} \theta| \leq 1/4$ into U and $a \mapsto g(a)$ is of class C^1 from B into $(\ell_\nu^1)^{18}$. Suppose that $\bar{A}_0$ and $\bar{A}_1$ are finite and that A_{fin} is invertible. Then T_ε maps B into X and is of class C^1 there, and $T_\varepsilon(x) = x$ if and only if x is a zero of $F(\cdot; \varepsilon)$. The derivative $DT_\varepsilon(\bar{x})$ has the finite rows of $I - A_{\text{fin}} DF_{\text{fin}}(\bar{x}; \varepsilon)$ and the tail rows $(DT_\varepsilon(\bar{x})y)_m = \sum_n A_m(\varepsilon) J'_n y_{m-n}$, $|m| > K$, with J'_n as in Section 4.4; it is the operator $I - A DF(\bar{x})$ of Section 4.5 with $d_m(\varepsilon)$ and $A_m(\varepsilon)$ in place of d_m and A_m . Finally, $DT_\varepsilon(x) - DT_\varepsilon(\bar{x})$ is obtained by applying $-A_{\text{fin}}$ and $-A_m(\varepsilon)$ to the rows of $DF(x; \varepsilon) - DF(\bar{x}; \varepsilon)$, which do not contain $d_m(\varepsilon)$.*

Proof. The finite part of T_ε involves the phase component, a bounded linear functional on X , and finitely many F_m , each a C^1 function of x on B . The tail part is the sum of $-(A_m(\varepsilon)i(\omega - \bar{\omega})ma_m)_m$, a bilinear map of $(\omega - \bar{\omega}, a)$ bounded because $|mA_m(\varepsilon)| \leq \bar{A}_1$ entrywise, and of the bounded linear maps $a \mapsto -(A_m(\varepsilon)\hat{J}_0 a_m)_m$ and $b \mapsto (A_m(\varepsilon)b_m)_m$, the latter applied to $g(a)$; so T_ε is of class C^1 from B into X . For $|m| > K$, the identity $A_m(\varepsilon)(i\bar{\omega}m - \hat{J}_0 + d_m(\varepsilon)E) = I$ gives $a_m - T_\varepsilon(x)_m = A_m(\varepsilon)F_m(x; \varepsilon)$, and $x_{\text{fin}} - T_\varepsilon(x)_{\text{fin}} = A_{\text{fin}}F_{\text{fin}}(x; \varepsilon)$; as A_{fin} and every $A_m(\varepsilon)$ are invertible, $T_\varepsilon(x) = x$ if and only if every component of $F(x; \varepsilon)$ vanishes. Differentiating the tail part at $\bar{x}$ in a direction y , with $\omega = \bar{\omega}$ and $\bar{a}_m = 0$ for $|m| > K$, gives $-A_m(\varepsilon)[\hat{J}_0 y_m - \sum_n J_n y_{m-n}] = \sum_n A_m(\varepsilon) J'_n y_{m-n}$. In $DF(x; \varepsilon) - DF(\bar{x}; \varepsilon)$ the term $d_m(\varepsilon)E y_m$ cancels, and the tail part of $DT_\varepsilon(x) - DT_\varepsilon(\bar{x})$ is $-A_m(\varepsilon)$ applied to $i(\omega - \bar{\omega})my_m + iy_\omega ma_m - [(Dg(a) - Dg(\bar{a}))y_a]_m$. $\square$

Lemma 4.11 (the derivative of d_m). $d'_m(\varepsilon) = 8\pi^4 Dm^4 S(w)S'(w)$ with $w = \pi^2 m^2 \varepsilon$ and $S'(w) = \sum_{k \geq 1} (-1)^k k w^{k-1} / (2k+1)!$, and $S'(w) = (\cos \sqrt{w} - S(w)) / (2w)$ for $w > 0$. If $0 \leq w \leq W$, $n \geq 1$, $q_n = W / ((2n+2)(2n+3)) < 1$ and $q'_n = W / (2n(2n+3)) < 1$, then $S(w)$ and $S'(w)$ differ from their partial sums over $k < n$ by at most $t_n / (1 - q_n)$ and $u_n / (1 - q'_n)$, with $t_n = W^n / (2n+1)!$ and $u_n = nW^{n-1} / (2n+1)!$.

Proof. The derivative is the chain rule applied to (4), and the closed form follows by differentiating $\sin \sigma / \sigma$, $\sigma = \sqrt{w}$. The k -th terms of the two series have moduli at most t_k and u_k , and for $k \geq n$, $t_{k+1}/t_k = W / ((2k+2)(2k+3)) \leq q_n$ and $u_{k+1}/u_k = W / (2k(2k+3)) \leq q'_n$, so the tails are dominated by geometric series. $\square$

The program covers P by subintervals with exact rational ends, evaluates S and S' on the corresponding balls of w in Arb (the partial sums with these remainders, or the closed forms where $w > 1$ is certain), and takes the union of the resulting enclosures of d'_m : a real ball DD_m that contains $d'_m(\xi)$ for every $\xi \in P$, for each $|m| \leq K$.

Lemma 4.12 (bounds uniform over a piece). *Let P be a piece and, for $|m| \leq K$, DD_m a real interval containing $d'_m(\xi)$ for every $\xi \in P$. For each output component c (the ω component or one of the 18 state components) let $Y_{0,c}^p$ be the bound of Section 4.5 for the component c of $AF(\bar{x})$, computed with $d_m(\varepsilon_c)$ in the finite rows and with the tail suprema over $|m| > K$ and $\varepsilon \in P$; let $Y_{0,c}^g$ bound $\|(A_{\text{fin}}w)_c\|_\nu$ for every w in the box W whose only nonzero entries are $W_{(V,m)} = DD_m \bar{a}_{V,m}$, $|m| \leq K$; let $B_{1,cc'} = \max(ff_{cc'}, ft_{cc'}) + \mathcal{T}_{cc'}$ be the block bounds of Section 4.5, computed with $d_m(\varepsilon_c)$ in the finite block and with the tail suprema over P ; and let*

$$B_{1,cV}^g = \max_{|m| \leq K} \frac{|DD_m|}{\nu^{|m|}} \sum_{(j,m') \in c} |(A_{\text{fin}})_{(j,m'),(V,m)}| \nu^{|m'|}, \quad B_{1,cc'}^g = 0 \quad (c' \neq V),$$

the sum over the rows of component c . Then for every $\varepsilon \in P$

$$\|T_\varepsilon(\bar{x}) - \bar{x}\|_\eta \leq Y_0 := \max_c \frac{Y_{0,c}^p + \delta_P Y_{0,c}^g}{\eta_c}, \quad \|DT_\varepsilon(\bar{x})\|_\eta \leq Z_1 := \max_c \frac{1}{\eta_c} \sum_{c'} \eta_{c'} (B_{1,cc'} + \delta_P B_{1,c'}^g).$$

Proof. An operator B on X with blocks $B_{cc'}$ has $\|B\|_\eta \leq \max_c \eta_c^{-1} \sum_{c'} \eta_{c'} \|B_{cc'}\|$, since $\|y_{c'}\| \leq \eta_{c'} \|y\|_\eta$. By the mean value theorem for the real function d_m , $d_m(\varepsilon) - d_m(\varepsilon_c) = (\varepsilon - \varepsilon_c) d'_m(\xi_m)$ for some $\xi_m \in P$, and $|\varepsilon - \varepsilon_c| \leq \delta_P$. In $F_{\text{fin}}(\bar{x}; \varepsilon)$ only the entries (V, m) contain the damping, so $F_{\text{fin}}(\bar{x}; \varepsilon) = F_{\text{fin}}(\bar{x}; \varepsilon_c) + (\varepsilon - \varepsilon_c)w$ with $w_{(V,m)} = d'_m(\xi_m) \bar{a}_{V,m}$, $w \in W$, and $\|(A_{\text{fin}} F_{\text{fin}}(\bar{x}; \varepsilon))_c\|_\nu \leq \|(A_{\text{fin}} F_{\text{fin}}(\bar{x}; \varepsilon_c))_c\|_\nu + \delta_P Y_{0,c}^g$. The tail components of $T_\varepsilon(\bar{x}) - \bar{x}$ are $A_m(\varepsilon)g_m(\bar{a})$, bounded as in Section 4.5 because the suprema are taken over $\varepsilon \in P$. For Z_1 , by Lemma 4.10: in the finite block, $I - A_{\text{fin}} J_{\text{fin}}(\varepsilon) = [I - A_{\text{fin}} J_{\text{fin}}(\varepsilon_c)] - A_{\text{fin}} \Delta(\varepsilon)$, where $\Delta(\varepsilon)$ is diagonal with the entries $d_m(\varepsilon) - d_m(\varepsilon_c)$ on the coordinates (V, m) and 0 elsewhere; column (V, m) of $A_{\text{fin}} \Delta(\varepsilon)$ is column (V, m) of A_{fin} times a number of modulus at most $\delta_P |\overline{DD_m}|$, so its weighted column sum over the rows of c , divided by $\nu^{|m|}$, is at most $\delta_P B_{1,cV}^g$. The finite rows of the tail columns contain no damping (it acts on the diagonal, in tail rows) and are bounded as in Section 4.5. The tail rows are $\sum_n A_m(\varepsilon) J'_n y_{m-n}$ and are bounded by $\mathcal{T}$ with the suprema over P . $\square$

Lemma 4.13 (Z_2 from a bound on the Hessian). *Let $R_1, \dots, R_{18} > 0$ and $\rho_2 > 1/4$ with $q_2 = \nu e^{-\rho_2} < 1$, $Q_2 = (1 + q_2)/(1 - q_2)$. Suppose that f is holomorphic on an open set containing the compact set $\mathcal{K} = \{\phi_{\bar{a}}(\theta) + w : |\text{Im } \theta| \leq \rho_2, |w_i| \leq R_i\}$ and that $|\partial_j \partial_l f_k| \leq MH_{kjl}$ on $\mathcal{K}$. Write $J_{kj} = \partial_j f_k$, $H_{kjl} = \partial_j \partial_l f_k$, and for $a \in (\ell_\nu^1)^{18}$ with $t_i := \|a_i - \bar{a}_i\|_\nu < R_i$ let $\Pi(t) = \prod_i R_i/(R_i - t_i)$. Then:*

- (a) ϕ_a maps the closed strip $|\text{Im } \theta| \leq 1/4$ into $\mathcal{K}$, $g(a)$, $J_{kj} \circ \phi_a$ and $H_{kjl} \circ \phi_a$ lie in ℓ_ν^1 , and $\|H_{kjl} \circ \phi_a\|_\nu \leq MH_{kjl} Q_2 \Pi(t)$;
- (b) $a \mapsto g(a)$ and $a \mapsto J_{kj} \circ \phi_a$ are of class C^1 into ℓ_ν^1 , with $(Dg(a)y)_k = \sum_j (J_{kj} \circ \phi_a) * y_j$ and $D(J_{kj} \circ \phi_a)h = \sum_l (H_{kjl} \circ \phi_a) * h_l$, where $*$ is the convolution product;
- (c) if $\eta_i r_* < R_i$ for every i , then for every $\varepsilon \in P$ and every x with $\|x - \bar{x}\|_\eta \leq r \leq r_*$, $\|DT_\varepsilon(x) - DT_\varepsilon(\bar{x})\|_\eta \leq Z_2 r$, where

$$Z_2 := \max_c \frac{1}{\eta_c} \sum_k [2\eta_\omega \eta_k (N_1 + \bar{A}_1)_{ck} + (N_0 + \bar{A}_0)_{ck} W_k], \quad W_k := Q_2 \Pi(\eta r_*) \sum_{j,l} \eta_j \eta_l MH_{kjl},$$

N_0 and N_1 are the block norms of Section 4.6, $\bar{A}_0$ and $\bar{A}_1$ the suprema over $|m| > K$ and $\varepsilon \in P$, and $\Pi(\eta r_*)$ is Π at $t_i = \eta_i r_*$; in the ω row only N_0 and N_1 enter.

Proof. (a) For $|\text{Im } \theta| \leq 1/4$, $|\phi_{a,i}(\theta) - \phi_{\bar{a},i}(\theta)| \leq t_i < R_i$, so $\phi_a(\theta) \in \mathcal{K}$. The rest is the proof of Lemma 4.7 with H_{kjl} , J_{kj} and f_k in place of f_k : for $|\text{Im } \theta| \leq \rho_2$ the Taylor coefficients $c_\alpha(\theta)$ of $w \mapsto H_{kjl}(\phi_{\bar{a}}(\theta) + w)$ satisfy $|c_\alpha(\theta)| \leq MH_{kjl} R^{-\alpha}$ by Cauchy's inequality on the closed polydisc of radii R , they are holomorphic and 2π -periodic in θ , and Lemma 4.2 gives $\|c_\alpha\|_\nu \leq MH_{kjl} R^{-\alpha} Q_2$; so with $h = a - \bar{a}$ the series $\sum_\alpha c_\alpha h^\alpha$ (convolution powers) converges absolutely in ℓ_ν^1 , with norm at most $MH_{kjl} Q_2 \prod_i (1 - t_i/R_i)^{-1}$, and its sum is $H_{kjl} \circ \phi_a$ because for real θ the Taylor series converges to $H_{kjl}(\phi_a(\theta))$ and convergence in ℓ_ν^1 is uniform convergence. For J_{kj} and f_k the same holds with bounds of $|J_{kj}|$ and $|f_k|$ on the compact set $\mathcal{K}$. (b) A power series in h with coefficients in the Banach algebra ℓ_ν^1 , dominated as above by a scalar power series in (t_i) with nonnegative coefficients that converges for $t_i < R_i$, defines a C^1 map on $\{t_i < R_i\}$ whose derivative is obtained by differentiating term by term, because the series of the derivatives of the terms is dominated by the derivatives of

the scalar series, which converge locally uniformly on the same set; the term-by-term derivative of the series of f_k (of J_{kj}) in the direction h_j (h_l) is the series of J_{kj} (H_{kjl}) applied to h_j (h_l), since the Taylor coefficients of $\partial_j f_k$ are those of f_k differentiated. (c) By Lemma 4.10, $DT_\varepsilon(x) - DT_\varepsilon(\bar{x})$ applies $-A$ to $(DF(x; \varepsilon) - DF(\bar{x}; \varepsilon))y = (im(y_\omega \Delta a_m + \Delta \omega y_m))_m - (Dg(\bar{a} + \Delta a) - Dg(\bar{a}))y_a$, with $\Delta = x - \bar{x}$. For $\|y\|_\eta \leq 1$, component k of $b = y_\omega \Delta a + \Delta \omega y_a$ has norm at most $|y_\omega| \|\Delta a_k\|_\nu + |\Delta \omega| \|y_k\|_\nu \leq 2\eta_\omega \eta_k r$, and $\|(A(imb))_c\| \leq \sum_k (N_1 + \bar{A}_1)_{ck} \|b_k\|_\nu$ as in Section 4.6. By (b) and the mean value inequality along the segment $\bar{a} + \sigma \Delta a$, $\sigma \in [0, 1]$, on which $t_i \leq \eta_i r \leq \eta_i r_* < R_i$,

$$\|[(Dg(\bar{a} + \Delta a) - Dg(\bar{a}))y_a]_k\|_\nu \leq \sum_{j,l} \sup_\sigma \|H_{kjl} \circ \phi_{\bar{a} + \sigma \Delta a}\|_\nu \|\Delta a_l\|_\nu \|y_j\|_\nu \leq r W_k,$$

because Π increases in each t_i ; applying A costs $(N_0 + \bar{A}_0)_{ck}$. Dividing by η_c gives the claim. $\square$

The program computes *MH* (`code/fourier/branch.py`, class `HessBound`) as the strip supremum of Lemma 4.1, on $|\operatorname{Im} \theta| \leq \rho_2 = 1$, of the second derivatives of f over the family of trigonometric polynomials obtained from $\phi_{\bar{a}}$ by replacing each constant coefficient $\bar{a}_{0,i}$ by the complex box of half-width $R_i = 32\eta_i r_*$ about it, a family that contains $\phi_{\bar{a}}(\theta) + w$ for $|w_i| \leq R_i$. The second derivatives are evaluated with second-order forward-mode dual numbers in Arb; every intermediate is checked to be finite and every logarithm and square-root argument to lie in $\{\operatorname{Re} w > 0\}$, so that, by the induction of Section 4.1 applied to the value, gradient and Hessian of each intermediate, a finite result encloses f , Df and $D^2 f$ and certifies that f is holomorphic on an open set containing $\mathcal{K}$.

Lemma 4.14 (continuity and gluing). (a) Let P be a piece and $0 < r_{\text{ex}} \leq r_{\text{un}} \leq r_*$ such that, for every $\varepsilon \in P$, T_ε maps $B_{r_{\text{ex}}}(\bar{x})$ into itself and is a κ -contraction on $B_{r_{\text{un}}}(\bar{x})$ (balls of $\|\cdot\|_\eta$), $\kappa < 1$, and let $x_*(\varepsilon)$ be its fixed point in $B_{r_{\text{ex}}}(\bar{x})$, which is its only fixed point in $B_{r_{\text{un}}}(\bar{x})$. Then $\varepsilon \mapsto x_*(\varepsilon)$ is continuous from P into X . (b) Let P_a and P_b be pieces as in (a) with $P_a \cap P_b \neq \emptyset$ and

$$\|\bar{x}_a - \bar{x}_b\|_{\eta(b)} + r_{\text{ex}}(a) \max_c \frac{\eta_c(a)}{\eta_c(b)} \leq r_{\text{un}}(b). \quad (10)$$

Then $x_{*,a} = x_{*,b}$ on $P_a \cap P_b$.

Proof. (a) For $\varepsilon, \varepsilon' \in P$ both fixed points lie in $B_{r_{\text{un}}}(\bar{x})$, where $T_{\varepsilon'}$ is a κ -contraction, so

$$\|x_*(\varepsilon') - x_*(\varepsilon)\|_\eta \leq \|T_{\varepsilon'}(x_*(\varepsilon')) - T_{\varepsilon'}(x_*(\varepsilon))\|_\eta + \|T_{\varepsilon'}(x) - T_\varepsilon(x)\|_\eta \leq \kappa \|x_*(\varepsilon') - x_*(\varepsilon)\|_\eta + \|T_{\varepsilon'}(x) - T_\varepsilon(x)\|_\eta$$

with $x = x_*(\varepsilon)$. It remains to show that $\varepsilon' \mapsto T_{\varepsilon'}(x)$ is continuous at ε for this fixed $x = (\omega, a)$. The finite part changes by A_{fin} applied to the finitely many terms $(d_m(\varepsilon') - d_m(\varepsilon))Ea_m$, $|m| \leq K$. The tail part changes by $-(A_m(\varepsilon') - A_m(\varepsilon))v_m$, $v_m = i(\omega - \bar{\omega})ma_m + \hat{J}_0 a_m - g_m(a)$; each term tends to 0, as $A_m(\cdot)$ is the inverse of a continuous family of invertible matrices, and entrywise $|(A_m(\varepsilon') - A_m(\varepsilon))v_m| \leq 2\bar{A}_1|\omega - \bar{\omega}||a_m| + 2\bar{A}_0|\hat{J}_0 a_m - g_m(a)|$, which is summable with the weights $\nu^{|m|}$; dominated convergence gives the claim. (b) Since $\|y\|_{\eta(b)} \leq \|y\|_{\eta(a)} \max_c \eta_c(a)/\eta_c(b)$, for $\varepsilon \in P_a \cap P_b$ the point $x_{*,a}(\varepsilon)$ lies in the closed ball of radius $r_{\text{un}}(b)$ about $\bar{x}_b$ in $\|\cdot\|_{\eta(b)}$. It is a zero of $F(\cdot; \varepsilon)$ by Lemma 4.10, hence a fixed point of the map T_ε of P_b in that ball, so it is $x_{*,b}(\varepsilon)$. $\square$

Proof of Theorem C. The computation. The program `code/fourier/alln.py` proves the 73 pieces of the record `data/fourier-existence-alln.json`. Each has width $1/4096$; consecutive pieces overlap by $1/32768$, the first starts at 0 and the last ends at $1/64$. For each piece it computes in Arb, with $K = 12$, $K' = 2K + 16 = 40$, $M = 128$ nodes, $\rho = 3/2$ and $\rho_2 = 1$ (the settings and the working precisions are stored in the record): the strip covers and the enclosures of Proposition 4.4

at the center, the balls DD_m of Lemma 4.11, the tail data of Lemma 4.9, the bounds Y_0 and Z_1 of Lemma 4.12, and the Hessian cover and Z_2 of Lemma 4.13 with $R_i = 32\eta_i r_*$; it checks $\vartheta < 1$, $\eta_i r_* < R_i$, $p(r) < 0$ and $Z_1 + Z_2 r < 1$ at $r = r_{\text{ex}}$ and at $r = r_{\text{un}} \leq r_*$, $\bar{\omega} - \eta_\omega r_{\text{ex}} > 0$ and $|\bar{a}_{1,V}| > \eta_V r_{\text{ex}}/\nu$, and stores the period interval $2\pi/[\bar{\omega} - \eta_\omega r_{\text{ex}}, \bar{\omega} + \eta_\omega r_{\text{ex}}]$ with outward-rounded decimal ends. Over the 73 pieces: Y_0 lies between 6.90×10^{-7} and 7.26×10^{-7} , almost all of it the term $\delta_P Y_0^g$ of the parameter width (the part at ε_c is at most 7.42×10^{-11}); Z_1 lies between 0.152 and 0.199, Z_2 between 1305 and 1373, r_{ex} between 8.14×10^{-7} and 9.07×10^{-7} , r_{un} between 4.65×10^{-4} and 6.11×10^{-4} , and $Z_1 + Z_2 r_{\text{un}} \leq 0.979$; $m_{\text{max}} = 391$ and $\vartheta < 0.4964$ on every piece; the weights lie in $[0.00216, 1]$, with $\eta_V = 1$; the period intervals are narrower than 1.24×10^{-5} ms.

Each piece. Fix a piece and $\varepsilon \in P$. By Lemma 4.13(a), (b), Lemma 4.10 applies on the ball $B_{r_*}(\bar{x})$ of $\|\cdot\|_\eta$ ($\|a_i - \bar{a}_i\|_\nu \leq \eta_i r_* < R_i$ there), once A_{fin} is known to be invertible; that follows from $Z_1 < 1$ as in Section 4.4, since the compression of $DT_\varepsilon(\bar{x})$ to the finite coordinates is $I - A_{\text{fin}} J_{\text{fin}}(\varepsilon)$ and the coordinate projection has norm 1 for $\|\cdot\|_\eta$. With Lemmas 4.12 and 4.13(c), the proof of Lemma 4.5, which uses only the three bounds and Banach's fixed point theorem, shows that T_ε maps $B_{r_{\text{ex}}}(\bar{x})$ into itself and is a contraction there and on $B_{r_{\text{un}}}(\bar{x})$, with constant at most $Z_1 + Z_2 r_{\text{un}} < 1$; so T_ε has exactly one fixed point $x_*(\varepsilon)$ in $B_{r_{\text{un}}}(\bar{x})$, it lies in $B_{r_{\text{ex}}}(\bar{x})$, and by Lemma 4.10 it is the only zero of $F(\cdot; \varepsilon)$ in $B_{r_{\text{un}}}(\bar{x})$. The proof of Lemma 4.8 applies to it with $\|\cdot\|_\eta$ (for which κ is an isometry), with $d_m(\varepsilon)$ real and even in m , with the domain certified by the Hessian cover (the points $\phi_{\bar{a}}(\theta) + h(\theta)$, θ real, $|h_j(\theta)| < R_j$), and with the checks $\bar{\omega} - \eta_\omega r_{\text{ex}} > 0$ and $|\bar{a}_{1,V}| > \eta_V r_{\text{ex}}/\nu$: $\omega_*(\varepsilon)$ is real and positive, $\phi_*(\cdot; \varepsilon)$ is real, analytic on $|\text{Im } \theta| < 1/4$ and of minimal period 2π , $\phi_{*,V}(0; \varepsilon) = s$, and $T(\varepsilon)$ lies in the stored interval. It solves (6) pointwise: the coefficient sequences of ϕ_* , ϕ'_* , ϕ''_* and $f(\phi_*)$ are absolutely summable (as $|m|^j \nu^{-|m|}$ is bounded), the shifts in $\mathcal{L}_\varepsilon$ act on mode m by $e^{\pm 2\pi i m h}$, and the vanishing of every $F_m(x_*; \varepsilon)$ is the equality of the Fourier coefficients of the two sides. By Lemma 4.14(a), x_* is continuous on P .

Gluing. The program checks that the lower ends and the upper ends of the pieces increase strictly and that $r_{\text{ex}} < r_{\text{un}}$ on each piece; only consecutive pieces overlap (the record lists no other overlap); and the 72 inequalities (10) for consecutive pieces hold in Arb, the largest ratio of the left side to $r_{\text{un}}(b)$ being below 0.0076. By Lemma 4.14(b) the pieces define one map x_* on $[0, 1/64]$. Every point of $[0, 1/64]$ has a neighborhood in $[0, 1/64]$ contained in one piece, because consecutive pieces overlap in intervals of positive length; so x_* is continuous. Since $\|\cdot\| \leq \|\cdot\|_\eta$, $x_*(\varepsilon)$ lies within $r_{\text{ex}} \leq 9.07 \times 10^{-7}$ of the center in the norm of Theorem B(a); since $r_{\text{un}} \min_c \eta_c \geq 1.05 \times 10^{-6}$ on every piece, the ball of that radius in that norm lies in $B_{r_{\text{un}}}(\bar{x})$, where $x_*(\varepsilon)$ is the only zero. This proves (a).

(b) Let $N \geq 8$, so $\varepsilon = 1/N^2 \in (0, 1/64]$ and $F(\cdot; \varepsilon)$ is the map (8) of the N -ring. The last part of Lemma 4.8, whose proof uses only that ϕ_* solves (2) and has minimal period 2π , gives (b) apart from the identification. For $N = 8, 16, 32, 64$ the program checks in Arb, for the piece P containing $1/N^2$ (for both pieces containing $1/4096$),

$$\|\bar{x}_E - \bar{x}_P\|_{\eta(P)} + r_{\text{ex}}^E \max_c \frac{1}{\eta_c(P)} \leq r_{\text{un}}(P), \quad (11)$$

where $\bar{x}_E$ and r_{ex}^E are the $K = 32$ center and the existence radius of Theorem B(a) (weights 1), and the difference of centers is formed in X with the coefficients beyond $K = 12$ of $\bar{x}_P$ equal to 0. Table 3 gives both sides. The zero of Theorem B(a) is a zero of $F(\cdot; 1/N^2)$ in the uniqueness ball of P , so it is $x_*(1/N^2)$, and Theorem B(c) applies to the wave.

(c) For $\varepsilon = 0$, $\phi_* = \phi_*(\cdot; 0)$ is real analytic and solves (6) with $\mathcal{L}_0 = 4\pi^2 D E \partial_\theta^2$, so by Section 2.3 $u(x, t) = \phi_*(\omega_* t + 2\pi x)$ is a real analytic solution of (7), 1-periodic in x . For each x , $t \mapsto u(x, t)$ has minimal period $2\pi/\omega_* = T(0)$, because ϕ_* has minimal period 2π , and $u(x, t + \tau') = u(x + \tau'/T(0), t)$,

so the profile goes once around the ring in each period. The interval for $T(0)$ is that of the piece $[0, 1/4096]$.

(d) By (a), x_* is continuous at 0, and $1/N^2 \rightarrow 0$. For $|\operatorname{Im} \theta| \leq 1/4$, $|\sum_m b_m e^{im\theta}| \leq \|b\|_\nu$, which gives uniform convergence of the profiles on the closed strip (in the scaled variables, hence also after the diagonal scaling Σ), and $T = 2\pi/\omega$ is continuous where $\omega > 0$. $\square$

The record also stores the inputs of every piece (exact centers, weights, r_* , the radii R_i , the run `log code/fourier/data/alln/pieces.jsonl`), so every bound can be recomputed; the tests `code/fourier/test_alln.py` re-prove five pieces from these inputs and compare every stored bound bit for bit; they passed in the run recorded in `review/alln-existence-fixcheck-2026-10-02.md`. The negative controls of the record are listed in Section 7.

5 Stability through one Hill operator

5.1 Setting

Let ϕ , ω , T , τ be as in Section 2.2, with ϕ solving (2), and let $A(\theta) = Df(\phi(\theta)) = \sum_n A_n e^{in\theta}$, a real analytic 2π -periodic matrix function, so $A_{-n} = \overline{A_n}$ and $\|A_n\|$ decays exponentially. The linearization along $x(t)$ is $y' = L(t)y$ with $(L(t)y)_j = A(\theta_j(t))y_j + cE(y_{j-1} - 2y_j + y_{j+1})$; let $Y(t)$ be its fundamental matrix with $Y(0) = I$, and $M_\tau = Q^{-1}Y(\tau)$, the derivative at $x(0)$ of $h = Q^{-1}\varphi_\tau$. The Floquet multipliers are the eigenvalues of $Y(T)$, with algebraic multiplicity. For a matrix M let $m(\lambda; M)$ be the algebraic multiplicity and $G_\lambda(M)$ the generalized eigenspace; for an operator H , $G_\mu(H) = \bigcup_k \ker(H - \mu)^k$ (with the convention of Appendix A on the domains of powers of H) and $m(\mu; H) = \dim G_\mu(H)$.

Let $|\cdot|_1$ be the ℓ^1 norm on $\mathbb{C}^{18}$, $\|B\|_{1 \rightarrow 1}$ the induced norm, $\mathcal{X} = \ell^1(\mathbb{Z}; \mathbb{C}^{18})$ with $\|P\| = \sum_m |P_m|_1$, and $\mathcal{D} = \{P \in \mathcal{X} : \sum_m |m| |P_m|_1 < \infty\}$. A sequence $P \in \mathcal{D}$ is the coefficient sequence of the C^1 function $p(\theta) = \sum_m P_m e^{im\theta}$. The Hill operator $H_0 : \mathcal{D} \rightarrow \mathcal{X}$ is

$$(H_0 P)_m = -i\omega m P_m + \sum_n A_n P_{m-n} - d_m E P_m, \quad (12)$$

the coefficient sequence of $\mathcal{L}p := -\omega p' + Ap + cE\Delta_N p$. The ansatz $y_j(t) = e^{\mu t} p(\theta_j(t))$ solves the linearization if and only if $\mathcal{L}p = \mu p$.

We use three standard facts about an operator H with compact resolvent on $\mathcal{X}$, or on the ℓ^1 space of Section 5.4, which Appendix A proves (Lemma A.2(c) and Theorem A.5; see also Kato's book [25, Ch. III, Sect. 6]): its spectrum consists of isolated eigenvalues of finite algebraic multiplicity; for a bounded open rectangle Ω whose boundary Γ lies in the resolvent set, the Riesz projection $\frac{1}{2\pi i} \oint_\Gamma (\mu - H)^{-1} d\mu$ (Γ counterclockwise) is a projection onto the sum of the generalized eigenspaces of the eigenvalues in Ω , so its rank is $n(H, \Omega) := \sum_{\mu \in \Omega} m(\mu; H)$; and the resolvent is holomorphic on the resolvent set.

Lemma 5.1. *H_0 is closed on $\mathcal{D}$ and has compact resolvent, and $\{\operatorname{Re} \mu > \alpha\}$ lies in its resolvent set, where $\alpha = \sum_n \|A_n\|_{1 \rightarrow 1} + 4c$.*

Proof. Write $H_0 = D_0 + B$ with $(D_0 P)_m = -i\omega m P_m$ on $\mathcal{D}$ and B the rest. By Young's inequality and $d_m \leq 4c$, $\|B\| \leq \alpha$. D_0 is closed (a diagonal operator on its maximal domain), and for $\lambda \notin i\omega\mathbb{Z}$, $(\lambda - D_0)^{-1}$ is diagonal with entries $(\lambda + i\omega m)^{-1}$, compact as the norm limit of its finite truncations. If $\operatorname{Re} \lambda > \alpha$, then $\|(\lambda - D_0)^{-1}\| \leq 1/\operatorname{Re} \lambda$, so $\|B(\lambda - D_0)^{-1}\| < 1$ and $\lambda - H_0 = (I - B(\lambda - D_0)^{-1})(\lambda - D_0)$ is a bijection $\mathcal{D} \rightarrow \mathcal{X}$ with compact inverse. H_0 is closed as a closed operator plus a bounded one, and compactness of the resolvent at one point gives it everywhere by the resolvent identity $R(\lambda) = R(\lambda_0)(I + (\lambda_0 - \lambda)R(\lambda))$. $\square$

5.2 The Hill-sector theorem

Lemma 5.2 (twisted periodicity). $L(t + \tau) = QL(t)Q^{-1}$, $Y(t + \tau) = QY(t)M_\tau$, $Y(k\tau) = Q^k M_\tau^k$ and $Y(T) = M_\tau^N$.

Proof. $(QL(t)Q^{-1}y)_j = A(\theta_{j+1}(t))y_j + cE(y_{j-1} - 2y_j + y_{j+1}) = (L(t + \tau)y)_j$, because $(Q^{-1}y)_k = y_{k-1}$ and $\theta_{j+1}(t) = \theta_j(t + \tau)$. So $Z(t) = Q^{-1}Y(t + \tau)$ solves $Z' = L(t)Z$ with $Z(0) = M_\tau$, and $Z(t) = Y(t)M_\tau$. Induction gives $Y(k\tau) = Q^k M_\tau^k$, and $Q^N = I$ gives $Y(T) = M_\tau^N$. $\square$

Theorem 5.3 (Hill sectors). *For every $\mu \in \mathbb{C}$, $m(\mu; H_0) = m(e^{\mu\tau}; M_\tau)$.*

Proof. Let $\lambda = e^{\mu\tau}$.

$m(\mu; H_0) \leq m(\lambda; M_\tau)$. Let $G = G_\mu(H_0)$, read through the Fourier correspondence as a finite-dimensional (Lemma 5.1 and Theorem A.5(iii)) $\mathcal{L}$ -invariant space of C^1 functions, with $\mathcal{L}|_G = \mu + K_G$, K_G nilpotent. For $p \in G$ put $u(t) = e^{t\mathcal{L}_G}p \in G$ and $y_j(t) = u(t)(\theta_j(t))$. The chain rule gives $y'_j = (\mathcal{L}_G u + \omega u')(\theta_j) = (Au + cE\Delta_N u)(\theta_j) = A(\theta_j)y_j + cE(y_{j-1} - 2y_j + y_{j+1})$, so y solves the linearization with $y(0) = \iota^*p$, $\iota^*p := (p(2\pi j/N))_{j \in \mathbb{Z}_N}$. As $\theta_j(t + \tau) = \theta_{j+1}(t)$, $y(\tau) = Q\iota^*(e^{\tau\mathcal{L}_G}p)$; also $y(\tau) = Y(\tau)\iota^*p$. Hence $M_\tau \iota^* = \iota^*e^{\tau\mathcal{L}_G}$ on G . The map ι^* is injective on G : if $\iota^*p = 0$ then $y = 0$, so for all t, j and integers n , $0 = u(t + nT)(\theta_j(t)) = e^{\mu(t+nT)} \sum_k \frac{(t+nT)^k}{k!} (K_G^k p)(\theta_j(t))$. For fixed t, j this is $e^{\mu(t+nT)}$ times a polynomial in n vanishing at every integer, so all its coefficients vanish; if $p \neq 0$ and k^* is the largest k with $K_G^k p \neq 0$, the coefficient of n^{k^*} is $\frac{T^{k^*}}{k^*!} (K_G^{k^*} p)(\theta_j(t))$, so $K_G^{k^*} p$ vanishes at every point, a contradiction. So $\iota^*(G)$ is an M_τ -invariant space of dimension $\dim G$ on which M_τ is similar to $\lambda e^{\tau K_G}$, whose only eigenvalue is λ ; hence $\iota^*(G) \subset G_\lambda(M_\tau)$.

$m(\mu; H_0) \geq m(\lambda; M_\tau)$. Let $W = G_\lambda(M_\tau)$, $r = \dim W \geq 1$, and on W write $M_\tau = \lambda(I + K)$, K nilpotent. Put $B = \mu I + \tau^{-1} \sum_{k=1}^r (-1)^{k+1} K^k / k$ on W . Then $e^{\tau B} = \lambda \exp(\log(I + K)) = M_\tau$ on W (an identity of formal power series, valid for nilpotent K), B commutes with $M_\tau|_W$, and the only eigenvalue of B is μ . Let $\Pi(t) = Y(t)e^{-tB} : W \rightarrow \mathbb{C}^{18N}$. By Lemma 5.2, $\Pi(t + \tau)w = QY(t)M_\tau e^{-\tau B} e^{-tB}w = Q\Pi(t)w$, so $\Pi_j(t)w = \Pi_0(t + j\tau)w$ and $\Pi_0(t + T)w = \Pi_0(t)w$. Hence $p_w(\theta) := \Pi_0(\theta/\omega)w$ is 2π -periodic, C^∞ , with coefficients in $\mathcal{D}$, and $\Pi_j(t)w = p_w(\theta_j(t))$. Differentiating $Y(t)w = \Pi(t)e^{tB}w$ gives $L(t)\Pi(t)w = \Pi'(t)w + \Pi(t)Bw$, whose component 0 at $t = \theta/\omega$ reads $\mathcal{L}p_w = p_{Bw}$. The map $w \mapsto p_w$ is injective ($p_w(2\pi j/N) = w_j$), so its image is an $\mathcal{L}$ -invariant space of dimension r on which $(\mathcal{L} - \mu)^r = 0$; it lies in $G_\mu(H_0)$. $\square$

Corollary 5.4. (i) $\text{spec } H_0 = \{\mu : e^{\mu\tau} \in \text{spec } M_\tau\}$, invariant with multiplicities under $\mu \mapsto \mu + i\omega N$ and $\mu \mapsto \bar{\mu}$. (ii) For every real a , on the half-open strip $S_a = \{a \leq \text{Im } \mu < a + \omega N\}$ the map $\mu \mapsto e^{\mu\tau}$ is a bijection from $\text{spec } H_0 \cap S_a$ onto $\text{spec } M_\tau$ preserving algebraic multiplicities, and these multiplicities add up to $18N$. (iii) For $\rho \neq 0$, $m(\rho; Y(T)) = \sum \{m(\mu; H_0) : \mu \in \text{spec } H_0 \cap S_a, e^{\mu T} = \rho\}$. (iv) Let H_q be H_0 with d_m replaced by d_{m+q} . Then H_q is similar to $H_0 + i\omega q$ by the shift $(SP)_k = P_{k-q}$, so the “ N sectors” H_q , $q \in \mathbb{Z}_N$, carry no information beyond H_0 .

Proof. (i) Theorem 5.3 in both directions (every logarithm of an eigenvalue of M_τ is an eigenvalue of H_0), $e^{i\omega N\tau} = 1$, and M_τ real. (ii) $e^{\mu\tau}$ has period $i\omega N$, so it is injective on S_a and onto $\mathbb{C} \setminus \{0\}$; M_τ is invertible. (iii) $Y(T) = M_\tau^N$; $\mathbb{C}^{18N}$ is the sum of the $G_\lambda(M_\tau)$, each invariant under M_τ^N with only eigenvalue $\lambda^N = e^{\mu T}$. (iv) Substitute $k = m + q$. $\square$

Remark 5.5 (half-open strips). The count $18N$ holds in a half-open strip of height ωN . A closed strip $|\text{Im } \mu| \leq \omega N/2$ can count an eigenvalue twice: M_τ is real, so a negative real eigenvalue of M_τ persists under perturbation and its logarithms lie exactly on $\text{Im } \mu \in \omega N/2 + \omega N\mathbb{Z}$. The certificate below counts eigenvalues only inside a rectangle whose boundary lies in the resolvent set.

Corollary 5.6 (section maps). *Let Σ be a C^1 hypersurface through $x^* = x(0)$ transverse to $F(x^*)$, and $g(z) = Q^{-1}\varphi_{t_g(z)}(z)$ a section map with t_g of class C^1 , $t_g(x^*) = \tau$, $g(\Sigma) \subset \Sigma$. Then $\det(\lambda I - M_\tau) = (\lambda - 1)\det(\lambda I - Dg(x^*))$.*

Proof. $Y(\tau)F(x^*) = F(x(\tau)) = QF(x^*)$, so $M_\tau F(x^*) = F(x^*)$. $Dg(x^*) = M_\tau + F(x^*)Dt_g(x^*)$ on $T\Sigma$ maps $T\Sigma$ into itself, so it is the compression of M_τ along $F(x^*)$; in the splitting span $F(x^*) \oplus T\Sigma$, M_τ is block upper triangular with diagonal blocks 1 and $Dg(x^*)$. $\square$

5.3 The trivial multiplier

Lemma 5.7. (a) $P^* = (ima_m)_m$, the coefficients of ϕ' , is a nonzero element of $\mathcal{D}$ with $H_0 P^* = 0$. (b) $m(1; Y(T)) = \sum_{q=0}^{N-1} m(i\omega q; H_0)$. (c) The multiplier 1 is algebraically simple if and only if $m(0; H_0) = 1$ and $i\omega q \notin \text{spec } H_0$ for $q = 1, \dots, N-1$.

Proof. (a) ϕ is analytic and nonconstant; differentiating (2) gives $\mathcal{L}\phi' = 0$. (b) Corollary 5.4(iii) with $\rho = 1$, $a = -\omega/2$: $e^{\mu T} = 1$ iff $\mu \in i\omega\mathbb{Z}$, and $i\omega q \in S_{-\omega/2}$ iff $q = 0, \dots, N-1$. (c) follows from (a) and (b). $\square$

An eigenvalue $i\omega q$, $q \not\equiv 0 \pmod{N}$, would be a neutral mode of the reduced map (a spatial phase twist) that merges with the multiplier 1 of $Y(T)$. The certificate below excludes it together with everything else in $\{\text{Re } \mu \geq -\delta\}$.

5.4 Exclusion and counting by a Riesz-projection homotopy

Goal: for a given $\delta > 0$, prove

$$\text{spec } H_0 \cap \{\text{Re } \mu \geq -\delta\} = i\omega N\mathbb{Z}, \quad \text{each eigenvalue algebraically simple.} \quad (13)$$

Lemma 5.8. *If P, P' are bounded projections on a Banach space, $\|P - P'\| < 1$ and $\text{rank } P < \infty$, then $\text{rank } P' = \text{rank } P$.*

Proof. If $x \in \text{Ran } P'$ and $Px = 0$ then $\|x\| = \|(P' - P)x\| < \|x\|$ unless $x = 0$; so P maps $\text{Ran } P'$ injectively into $\text{Ran } P$. Exchange the roles. $\square$

Lemma 5.9 (homotopy count). *Let X be one of the spaces $\ell_w^1(\mathcal{J})$ of Appendix A, $\hat{D}$ an operator in X with compact resolvent, $\hat{E} \in B(X)$, Ω a bounded open rectangle whose boundary Γ lies in the resolvent set of $\hat{D}$, and suppose $I - s\hat{E}(\mu - \hat{D})^{-1}$ is invertible for all $(s, \mu) \in [0, 1] \times \Gamma$. Then, with $H(s) = \hat{D} + s\hat{E}$, Γ lies in the resolvent set of every $H(s)$ and $n(H(1), \Omega) = n(\hat{D}, \Omega)$.*

Proof. This is Theorem A.7. $\square$

The comparison operator. The program fixes:

- cell coordinates: an exact diagonal $S = \text{diag}(2^{k_1}, \dots, 2^{k_{18}})$, applied in every mode (its exponents k_l are chosen per N and stored in the records; they are not the scale exponents e_i of Section 2.1). This is a similarity; it commutes with E , leaves $-i\omega m$ and d_m unchanged, and changes no spectrum or multiplicity. From here on A_n and every norm are taken in these coordinates;
- a window $W = \{|m| \leq K_e\}$ ($n_W = 18(2K_e + 1)$ coordinates) and the tail $\text{Tl} = \{|m| > K_e\}$;
- an invertible complex $n_W \times n_W$ matrix V and a diagonal $\Lambda = \text{diag}(\lambda_j)$ with exactly representable entries (floating-point eigenvectors and eigenvalues of the window block);

- an exact real 18×18 matrix A_0^c (the midpoint of the enclosure of A_0), and for each residue $r \in \{0, \dots, \lfloor N/2 \rfloor\}$ an invertible U_r and an exact diagonal Λ_r approximately diagonalizing $X_r := A_0^c - d_r E$;
- the rectangle $\Omega = (-\delta, R_0) \times (a, b)$ (real part, imaginary part), with boundary Γ .

Let Scal be V on the window coordinates and the identity on the tail; it is bounded, boundedly invertible and maps $\mathcal{D}$ onto $\mathcal{D}$. Define

$$\hat{D} = \Lambda \oplus \bigoplus_{m \in \text{Tl}} B_m, \quad B_m = -i\omega m I + X_{m \bmod N}, \quad \hat{E} = \text{Scal}^{-1} H_0 \text{Scal} - \hat{D}.$$

The domain of $\hat{D}$ is $D(\hat{D}) = \{v : \sum_{m \in \text{Tl}} |m| |v_m|_1 < \infty\}$, the window coordinates being unrestricted; it is the maximal domain of the block-diagonal $\hat{D}$, because $(\omega|m| - \|X_r\|_{1 \rightarrow 1})|v_m|_1 \leq |B_m v_m|_1 \leq (\omega|m| + \|X_r\|_{1 \rightarrow 1})|v_m|_1$ with $r = m \bmod N$. Since Scal acts only on the finitely many window coordinates, $\text{Scal}^{-1}(\mathcal{D}) = D(\hat{D})$. In blocks, $\hat{E}_{WW} = R_W := V^{-1} H_{WW} V - \Lambda$, $\hat{E}_{WT} = V^{-1} H_{WT}$, $\hat{E}_{TW} = H_{TW} V$ and $(\hat{E}_{TT} P)_m = \sum_{n \neq 0, m-n \in \text{Tl}} A_n P_{m-n} + (A_0 - A_0^c) P_m$, where H_{WW} , H_{WT} , H_{TW} are the blocks of H_0 ; in the tail diagonal the damping $-d_m E$ of H_0 cancels exactly against the $-d_{m \bmod N} E$ in $X_{m \bmod N}$, by (3), and $-i\omega m$ cancels against B_m . These formulas define $\hat{E}$ as a bounded operator on the whole space (R_W is a finite matrix, and the other blocks have column sums bounded by $\sum_n \|A_n\|_{1 \rightarrow 1}$ times constants), and on $D(\hat{D})$ it equals $\text{Scal}^{-1} H_0 \text{Scal} - \hat{D}$; so $\hat{D} + \hat{E} = \text{Scal}^{-1} H_0 \text{Scal}$ as operators with the same domain $D(\hat{D})$. The exact ω stays in B_m : an error $(\omega - \bar{\omega})m$ is unbounded in m and cannot be moved into $\hat{E}$. The norm on the Scal -coordinates v is $\|v\| = \sum_j |v_j| + \sum_{m \in \text{Tl}} |v_m|_1$ (all weights equal to 1 in the runs), and the norm of an operator is the supremum of its column sums.

Notation for the bounds. All are upper bounds unless compared from below:

- dist_j : the distance from λ_j to Γ (a lower bound);
- fm_j : the ℓ^1 norm of column j of R_W ; $r_j = \sum_{w \in W} t_w |V_{(w, \cdot), j}|_1$ with $t_w = \sum_{|w+n| > K_e} \|A_n\|_{1 \rightarrow 1}$;
- $b_m = \max_k \sum_j |(V^{-1} H_{W, m})_{j, k}|$ for $m \in \text{Tl}$, where $H_{W, m}$ has entries $(A_{w-m})_{l, k}$ in row (w, l) ; $\hat{b} = \sup_{m \in \text{Tl}} b_m$;
- $\sigma_{\text{off}} = \sum_{n \neq 0} \|A_n\|_{1 \rightarrow 1}$ and $\theta_c = \sigma_{\text{off}} + \|A_0 - A_0^c\|_{1 \rightarrow 1}$;
- ρ_T : a bound of $\|(\mu - B_m)^{-1}\|_{1 \rightarrow 1}$ over $\mu \in \bar{\Omega}$ and $m \in \text{Tl}$ (Lemma 5.10), and $\theta_T = \theta_c \rho_T$.

Lemma 5.10 (tail blocks). *Let $h = \max(|a|, |b|)$, $\omega_{\text{lo}} \leq \omega$, $g_0 = \omega_{\text{lo}}(K_e + 1) - h$, $\eta > 0$, and*

$$\gamma_r = \min_l \max(-\delta - \eta - \text{Re } \lambda_{r, l}, g_0 - \eta - |\text{Im } \lambda_{r, l}|), \quad F_r = U_r^{-1} X_r U_r - \Lambda_r, \quad \kappa_r \geq \|U_r\|_{1 \rightarrow 1} \|U_r^{-1}\|_{1 \rightarrow 1}.$$

If $\gamma_r > \|F_r\|_{1 \rightarrow 1}$ for every r , then for every μ in the open neighborhood $N_\eta = \{-\delta - \eta < \text{Re } \mu < R_0 + \eta, |\text{Im } \mu| < h + \eta\}$ of $\bar{\Omega}$ and every $m \in \text{Tl}$, $\mu - B_m$ is invertible with $\|(\mu - B_m)^{-1}\|_{1 \rightarrow 1} \leq \rho_T := \max_r \kappa_r / (\gamma_r - \|F_r\|_{1 \rightarrow 1})$. Moreover, for fixed μ , $\|(\mu - B_m)^{-1}\|_{1 \rightarrow 1} \rightarrow 0$ as $|m| \rightarrow \infty$.

Proof. $\mu - B_m = z - X_r$ with $z = \mu + i\omega m$ and $r = m \bmod N$ (or $N - r$, which gives the same X_r). For $m \geq K_e + 1$, $\text{Re } z = \text{Re } \mu > -\delta - \eta$ and $\text{Im } z \geq \omega_{\text{lo}}(K_e + 1) - h - \eta = g_0 - \eta$; for $m \leq -(K_e + 1)$ the same holds for $\bar{z}$, and since X_r is real, $(\bar{z} - X_r)^{-1} = (z - X_r)^{-1}$ has the same norm. For such z , $|z - \lambda_{r, l}| \geq \max(-\delta - \eta - \text{Re } \lambda_{r, l}, g_0 - \eta - |\text{Im } \lambda_{r, l}|) \geq \gamma_r$, so $\|(z - \lambda_r)^{-1}\| \leq 1/\gamma_r$, and $z - X_r = U_r(z - \Lambda_r - F_r)U_r^{-1}$ with the Neumann series gives the bound. The decay follows from the Neumann bound $1/(\|\mu + i\omega m\| - \|X_r\|)$. $\square$

Lemma 5.11 (comparison operator). *If the hypothesis of Lemma 5.10 holds and $\text{dist}_j > 0$ for every j , then $\hat{D}$ has compact resolvent, Γ lies in its resolvent set, $n(\hat{D}, \Omega) = \#\{j : \lambda_j \in \Omega\}$, and $\|(\mu - \hat{D})^{-1}\| \leq \max(\max_j 1/\text{dist}_j, \rho_T)$ on Γ .*

Proof. $\hat{D}$ is block diagonal. Let D_T be the tail part, with domain $D(D_T) = \{v : \sum_{m \in \text{Tl}} |m| |v_m|_1 < \infty\}$ in the tail space. For $\mu \in N_\eta$, Lemma 5.10 gives $\|(\mu - B_m)^{-1}\|_{1 \rightarrow 1} \leq \rho_T$ for every $m \in \text{Tl}$, and $B_m(\mu - B_m)^{-1} = \mu(\mu - B_m)^{-1} - I$ has norm at most $1 + |\mu|\rho_T$. So the block-diagonal operator with blocks $(\mu - B_m)^{-1}$ maps the tail space into $D(D_T)$ (by the lower bound on $|B_m v_m|_1$ above), it is inverse to $\mu - D_T$ on both sides, and its norm, the supremum of the $1 \rightarrow 1$ norms of its blocks, is at most ρ_T : $\mu - D_T$ is a bijection of $D(D_T)$ onto the tail space with inverse of norm at most ρ_T . For fixed μ the blocks of the inverse tend to 0 in norm, so its finite truncations converge to it in norm, and it is compact. The resolvent of D_T is holomorphic on $N_\eta \supset \bar{\Omega}$, so its contour integral over Γ vanishes (Lemmas A.2(c) and A.3(d)), and the Riesz projection of $\hat{D}$ for Ω is the diagonal projection onto the window coordinates with $\lambda_j \in \Omega$; its rank is $n(\hat{D}, \Omega)$ by Theorem A.5(iv). $\square$

Lemma 5.12 (small gain, Schur-complement form). *Under the hypotheses of Lemma 5.11, if $\theta_T < 1$, $\hat{b} < \infty$ and*

$$fm_j + \frac{\hat{b} \rho_T r_j}{1 - \theta_T} < \text{dist}_j \quad \text{for every window column } j, \quad (\text{SC})$$

then $I - s\hat{E}(\mu - \hat{D})^{-1}$ is invertible for every $(s, \mu) \in [0, 1] \times \Gamma$.

Proof. Fix (s, μ) . The tail block of $\mu - \hat{D} - s\hat{E}$ is $(I - s\hat{E}_{TT}(\mu - D_T)^{-1})(\mu - D_T)$, invertible with $\|(\mu - D_T - s\hat{E}_{TT})^{-1}\| \leq \rho_T/(1 - \theta_T)$, since $\|\hat{E}_{TT}(\mu - D_T)^{-1}\| \leq \theta_c \rho_T$ (the column sums of $\hat{E}_{TT}$ are at most $\sum_{n \neq 0} \|A_n\| + \|A_0 - A_0^c\|$). By the block factorization (valid because the window block is finite-dimensional and both triangular factors are bijections of the domain), $\mu - \hat{D} - s\hat{E}$ is invertible if and only if the Schur complement

$$\text{Sc} = \mu - \Lambda - sR_W - s^2\hat{E}_{WT}(\mu - D_T - s\hat{E}_{TT})^{-1}\hat{E}_{TW} = (I - C_c(\mu - \Lambda)^{-1})(\mu - \Lambda)$$

is invertible, with $C_c = sR_W + s^2\hat{E}_{WT}(\cdots)^{-1}\hat{E}_{TW}$. Column j of $\hat{E}_{TW}$ has norm at most r_j (its tail rows are $\sum_w A_{m-w} V_{(w, \cdot), j}$, and $\sum_{m \in \text{Tl}} \|A_{m-w}\| \leq t_w$), and $\|\hat{E}_{WT}u\| \leq \sum_{m,k} |u_{m,k}| b_m \leq \hat{b}\|u\|$. So column j of C_c has norm at most $fm_j + \hat{b}\rho_T r_j/(1 - \theta_T)$, and $\|C_c(\mu - \Lambda)^{-1}\| \leq \max_j (fm_j + \hat{b}\rho_T r_j/(1 - \theta_T))/\text{dist}_j < 1$. Then $\mu - \hat{D} - s\hat{E}$ is invertible, which is equivalent to the claim because $\mu - \hat{D}$ is. $\square$

The condition (SC) asks only for the product of the window-to-tail coupling $\hat{b}$ and the tail-to-window coupling r_j of each column to be small, which suits the problem: eigenvectors of near-axis eigenvalues are small at the window edge. The programs also evaluate the stronger form in which the window and tail columns are bounded separately; it does not hold at any N and is not used.

Lemma 5.13 (coupling sums from coefficient bounds). *Suppose enclosures $[A_n]$ of A_n are known for $|n| \leq n_A$ and $\|A_n\|_{1 \rightarrow 1} \leq s_1 q_1^{|n|} + s_2 q_2^{|n|}$ for every n ($q_1, q_2 < 1$). Let $G_{\text{tail}}(k) = 2 \sum_i s_i q_i^k / (1 - q_i)$. Then $\sigma_{\text{off}} \leq \sum_{0 < |n| \leq n_A} \|[A_n]\| + G_{\text{tail}}(n_A + 1)$; $t_w \leq \sum_{|n| \leq n_A, |w+n| > K_e} \|[A_n]\| + G_{\text{tail}}(n_A + 1)$; b_m for $K_e < |m| \leq K_e + n_c$ is bounded from entrywise enclosures of A_{w-m} (for $|w-m| > n_A$ the entrywise tail enclosures of Section 5.5); and for $|m| > K_e + n_c$, $b_m \leq \beta_{\text{max}} G_{\text{tail}}(|m| - K_e)/2$, a bound that decreases in $|m|$, where $\beta_{\text{max}} = \max_{(w,l)} \sum_j |(V^{-1})_{j,(w,l)}|$.*

Proof. Sums of upper bounds, with exact geometric sums. For the last bound,

$$\sum_{(w,l)} \beta_{(w,l)} |(A_{w-m})_{l,k}| \leq \beta_{\max} \sum_w \|A_{w-m}\|_{1 \rightarrow 1},$$

and the indices $n = w - m$ have one sign with $|n| \geq |m| - K_e$. $\square$

Lemma 5.14 (the window without inverting V). *Let V_i be an exact floating-point approximate inverse of V , $C = I - V_i V$, and suppose $q_C := \|C\| < 1$. Then V is invertible, $V^{-1} = (I - C)^{-1} V_i$ with $\|(I - C)^{-1}\| \leq 1/(1 - q_C)$, and with $\Xi := V_i H_{WW} V - \Lambda$,*

$$fm_j \leq \frac{\|\Xi e_j\| + |\lambda_j| \|C e_j\|}{1 - q_C}, \quad \sum_j |(V^{-1})_{j,(w,l)}| \leq \frac{\|V_i e_{(w,l)}\|}{1 - q_C}.$$

Proof. $V_i V = I - C$ is invertible by the Neumann series, so V is invertible and $V^{-1} = (V_i V)^{-1} V_i$. Since $V_i V \Lambda = \Lambda - C \Lambda$, $V_i (H_{WW} V - V \Lambda) = \Xi + C \Lambda$ exactly, so $R_W = V^{-1} (H_{WW} V - V \Lambda) = (I - C)^{-1} (\Xi + C \Lambda)$, and $C \Lambda e_j = \lambda_j C e_j$. $\square$

Theorem 5.15 (the certificate). *Suppose the following are checked in ball arithmetic:*

- (C0) $\omega \in [\omega_{\text{lo}}, \omega_{\text{hi}}]$, $\omega_{\text{lo}} > 0$; entrywise enclosures $[A_n]$ of the true A_n for every n that enters (those of Section 5.5, for $|n| \leq n_A$ and beyond) and the tail form of Lemma 5.13; balls containing d_m ;
- (C1) $\delta > 0$; $R_0 > \alpha^{\text{up}} \geq \alpha$; $a < 0 < b$, $b - a \geq \omega_{\text{hi}} N$, $b < \omega_{\text{lo}} N$, $-a < \omega_{\text{lo}} N$;
- (C2) the hypothesis of Lemma 5.10 for $r = 0, \dots, \lfloor N/2 \rfloor$;
- (C3) V invertible and fm_j bounded from a ball matrix containing the true H_{WW} (with the ball $[\omega_{\text{lo}}, \omega_{\text{hi}}]$ on its diagonal); $\text{dist}_j > 0$ for every j ;
- (C4) $\theta_T < 1$ and (SC);
- (C5) exactly one λ_j lies in Ω .

Then (13) holds.

Proof. By Lemmas 5.11 and 5.12, Lemma 5.9 applies to $\hat{D}$ and $\hat{E}$, so $n(\text{Scal}^{-1} H_0 \text{Scal}, \Omega) = n(\hat{D}, \Omega) = 1$ and Γ lies in the resolvent set of H_0 ; the similarity by Scal gives $n(H_0, \Omega) = 1$. By (C1), $0 \in \Omega$, and 0 is an eigenvalue (Lemma 5.7(a)), so it is the only eigenvalue in Ω and $m(0; H_0) = 1$. Let μ be an eigenvalue with $\text{Re } \mu \geq -\delta$. By Lemma 5.1, $\text{Re } \mu \leq \alpha < R_0$. Since $b - a \geq \omega_{\text{hi}} N \geq \omega N$, some $\mu' = \mu + i\omega N k$ has $a \leq \text{Im } \mu' < a + \omega N \leq b$, and μ' is an eigenvalue of the same multiplicity (Corollary 5.4(i)). As Γ is in the resolvent set, $\mu' \in \Omega$, so $\mu' = 0$, $\mu \in i\omega N \mathbb{Z}$ and $m(\mu; H_0) = 1$. $\square$

The conditions $b < \omega_{\text{lo}} N$ and $-a < \omega_{\text{lo}} N$ are not needed for soundness (if Ω contained $\pm i\omega N$, the certificate could not pass, since by Theorem 5.15's argument Ω would contain at least two eigenvalues of H_0); they are needed for a correct spectrum to pass.

5.5 From the existence ball to the coefficients

Theorem 5.15 needs enclosures of the coefficients A_n of $Df(\phi_*(\theta))$ for the true profile, which is known only to lie in the ball of Theorem B(a). Let J_n be the coefficients of $Df(\phi_{\bar{a}}(\theta))$ for the center, enclosed as in Section 4.1 with the entrywise strip bound $|J_{n,kl}| \leq S_{J,kl}e^{-\rho|n|}$, and let M_k and R_j be as in Section 4.6.

Lemma 5.16. *Let $r = r_{\text{ex}}$, $\rho_e = \min(1/4, \rho_2) = 1/4$, $t_j = r$, assume $t_j < R_j$ for every j , and let*

$$\varepsilon_{kl} = \frac{M_k}{R_l} \left[\left(1 - \frac{t_l}{R_l}\right)^{-1} \prod_j \left(1 - \frac{t_j}{R_j}\right)^{-1} - 1 \right].$$

Then $|A_{kl}(\theta) - J_{kl}(\theta)| \leq \varepsilon_{kl}$ for $|\text{Im } \theta| \leq \rho_e$, and $|(A_n - J_n)_{kl}| \leq \varepsilon_{kl}e^{-\rho_e|n|}$ for every n .

Proof. For $|\text{Im } \theta| \leq 1/4$, $|\phi_{*,j}(\theta) - \phi_{\bar{a},j}(\theta)| \leq \|a_{*,j} - \bar{a}_j\|_\nu \leq t_j$, so $w = \phi_* - \phi_{\bar{a}}$ lies in the polydisc of radii t . On the polydisc of radii R , $f_k(\phi_{\bar{a}}(\theta) + w) = \sum_\alpha c_\alpha w^\alpha$ with $|c_\alpha| \leq M_k R^{-\alpha}$, so $\partial_l f_k(\phi_{\bar{a}} + w) - \partial_l f_k(\phi_{\bar{a}}) = \sum_{\alpha_l \geq 1, |\alpha| \geq 2} \alpha_l c_\alpha w^{\alpha - e_l}$ is bounded by the same sum for the majorant $M_k \prod_j (1 - t_j/R_j)^{-1}$, which is ε_{kl} . $A - J$ is holomorphic and 2π -periodic on $|\text{Im } \theta| < 1/4$; Lemma 4.2 on every strip $|\text{Im } \theta| \leq \rho''$, $\rho'' < 1/4$, and the limit $\rho'' \rightarrow 1/4$ give the coefficient bound. $\square$

In the cell coordinates S every bound transforms exactly $((S^{-1}MS)_{kl} = M_{kl}2^{k_l - k_k})$, and from here on $[J_n]$, S_J and ε denote the transformed bounds. The radius must be r_{ex} , not r_{un} : at $N = 8$ the record gives $\|\varepsilon\|_{1 \rightarrow 1} \leq 1.2 \times 10^{-12}$ in the coordinates S with r_{ex} ; since ε grows linearly in r for small r , the radius $r_{\text{un}} = 10^{-12}$ would give about 7×10^3 in the same coordinates, far above the near-axis margins $\text{dist}_j \approx 1.3 \times 10^{-6}$. So the program uses the entrywise enclosures $[A_n] = [J_n] + \text{ball}(0, \varepsilon e^{-\rho_e|n|})$ for $|n| \leq n_A = 80$ and $[A_n] = \text{ball}(0, S_J e^{-\rho|n|} + \varepsilon e^{-\rho_e|n|})$ for $|n| > 80$ (entrywise; this holds for every n by Lemma 4.2 applied to J on the strip $|\text{Im } \theta| \leq \rho$ and by Lemma 5.16), which is needed in the window ($|n| \leq 2K_e$, up to 96 at $N = 64$) and for b_m ($|n| \leq 2K_e + n_c$, up to 120), together with the tail form $\|A_n\|_{1 \rightarrow 1} \leq \|S_J\|e^{-\rho|n|} + \|\varepsilon\|e^{-\rho_e|n|}$ for every n , which is the form of Lemma 5.13.

5.6 Conclusion for the wave

Theorem 5.17. *Assume the existence statement of Lemma 4.8 and the certificate of Theorem 5.15 for some $\delta > 0$, with inputs (C0) built as in Section 5.5. Then:*

- (i) *$Y(T)$ has the eigenvalue 1 algebraically simple, and the other $18N - 1$ Floquet multipliers satisfy $|\rho| < e^{-\delta T} \leq e^{-\delta T_{\text{lo}}}$, $T_{\text{lo}} = 2\pi/\omega_{\text{hi}}$;*
- (ii) *M_τ has the eigenvalue 1 algebraically simple and all others of modulus less than $e^{-\delta\tau}$, and so has every section map of Corollary 5.6, without the eigenvalue 1;*
- (iii) *the orbit is locally exponentially orbitally stable with asymptotic phase, as stated in Theorem B(c).*

Proof. (i) By Corollary 5.4(iii) with $a = -\omega/2$, the multipliers are $e^{\mu T}$, $\mu \in \text{spec } H_0 \cap S_a$, with multiplicities. By (13), $\mu = 0$ is simple and is the only element of $i\omega\mathbb{Z} \cap S_a$ in the spectrum, and every other μ has $\text{Re } \mu < -\delta$, so $|e^{\mu T}| < e^{-\delta T}$. Lemma 5.7(b) gives simplicity of the multiplier 1. (ii) Corollaries 5.4(ii) and 5.6.

(iii) F is analytic near $O = \{x(t)\}$. Let $\Sigma = \{z : \langle F(x^*), z - x^* \rangle = 0\}$. Since $\frac{d}{dt} \langle F(x^*), \varphi_t(z) - x^* \rangle$ at (T, x^*) equals $|F(x^*)|^2 > 0$, the implicit function theorem gives a neighborhood U of x^* and a C^1 function t_S on U with $t_S(x^*) = T$ and $\varphi_{t_S(z)}(z) \in \Sigma$. The Poincaré map $P(z) = \varphi_{t_S(z)}(z)$ on $\Sigma \cap U$

has $P(x^*) = x^*$ and, as in Corollary 5.6 with Q replaced by I and τ by T , $DP(x^*)$ is the compression of $Y(T)$, with spectrum that of $Y(T)$ minus one copy of 1; by (i) its spectral radius ϱ is less than $e^{-\delta T}$. Let $\delta' < \delta$, $\kappa = e^{-\delta' T}$, and choose κ' with $\varrho < \kappa' < \kappa$. With a Jordan form $J = U_0^{-1} DP(x^*) U_0$ and $D_\epsilon = \text{diag}(1, \epsilon_J, \epsilon_J^2, \dots)$, the norm $|v|_* = |D_\epsilon^{-1} U_0^{-1} v|_\infty$ with $0 < \epsilon_J \leq \kappa' - \varrho$ gives $\|DP(x^*)\|_* \leq \kappa'$. By continuity there is $\epsilon_1 > 0$ with $\|DP(z)\|_* \leq \kappa$ on the ball $B_c = \{z \in \Sigma : |z - x^*|_* \leq \epsilon_1\}$, which is convex; the mean value inequality gives $|P(z) - x^*|_* \leq \kappa|z - x^*|_*$ there, so P maps B_c into itself and $|P^k(z) - x^*|_* \leq \kappa^k|z - x^*|_*$.

Let x^0 be given and choose $s \in [0, T]$ with $|x^0 - x(s)| = \text{dist}(x^0, O)$ (it exists because O is compact), and $z_0 = \varphi_{T-s}(x^0)$; by Gronwall $|z_0 - x^*| \leq e^{\text{Lip } T} |x^0 - x(s)|$, so the solution meets Σ at time $t_1 = T - s + t_S(z_0) \leq 3T$ at $z_1 = \varphi_{t_S(z_0)}(z_0)$ with $|z_1 - x^*| \leq C_1 \text{dist}(x^0, O)$, and we take ϵ_0 so small that $z_1 \in B_c$. Let $z_k = P^{k-1}(z_1)$ and $t_{k+1} = t_k + t_S(z_k)$. Since $|t_S(z_k) - T| \leq C_2 \kappa^k |z_1 - x^*|$, the limit $\sigma_\infty = \lim(t_k - kT)$ exists with $|t_k - kT - \sigma_\infty| \leq C_3 \kappa^k |z_1 - x^*|$. Put $\sigma = -\sigma_\infty$. For $t \in [t_k, t_{k+1}]$, $\varphi_t(x^0) = \varphi_{t-t_k}(z_k)$ and $x(t + \sigma) = \varphi_{t-t_k}(x(t_k + \sigma))$, where $x(t_k + \sigma) = x(t_k - kT - \sigma_\infty)$ lies within $\sup |F| C_3 \kappa^k |z_1 - x^*|$ of x^* and $|z_k - x^*| \leq C_4 \kappa^k |z_1 - x^*|$; Gronwall over an interval of length at most $2T$ gives $|\varphi_t(x^0) - x(t + \sigma)| \leq C_5 \kappa^k |z_1 - x^*|$, and $t \leq (k+1)T + C_6$ gives $\kappa^k \leq C_7 e^{-\delta' t}$. On $[0, t_1]$, Gronwall gives $|\varphi_t(x^0) - x(t + s)| \leq e^{\text{Lip } t} |x^0 - x(s)|$, and since $|\sigma_\infty - (t_1 - T)| \leq \sum_{k \geq 1} |t_S(z_k) - T| \leq C' |z_1 - x^*|$ and $t_1 - T = -s + t_S(z_0)$ with $|t_S(z_0) - T| \leq C |z_0 - x^*|$, $\sigma = s - T + O(\text{dist}(x^0, O))$, so by T -periodicity $|x(t + \sigma) - x(t + s)| \leq C'' \text{dist}(x^0, O)$. Together, on $[0, t_1]$ the difference is at most $C_8 e^{3\delta' T} e^{-\delta' t} \text{dist}(x^0, O)$. $\square$

This is the classical Andronov–Witt statement, proved here from the Poincaré map so that no textbook hypothesis needs matching. If T were not the minimal period of the ring solution, (i) to (iii) would still hold with respect to T ; Lemma 4.8 shows that it is minimal.

5.7 The computations

The program `code/fourier/stability.py` implements Theorem 5.15 with Lemma 5.10 for (C2), Lemma 5.12 for (C4) and Lemma 5.14 for (C3), at 128 bits everywhere. It reads ω_{lo} , ω_{hi} , r_{ex} and the settings from the Stage E record (it refuses to run if any source hash in that record differs from the current file, and it stores the SHA-256 of the record it read), and recomputes $[J_n]$, S_J and M_k by the same calls as the existence program; as a consistency check their suprema and the digests of the trigonometric polynomials must equal the values in the Stage E record. It asserts $t_j < R_j$; rounds δ up to a dyadic with denominator 2^{60} ; takes $R_0 = 32$, the least power of two above α^{up} , which lies between 17.84 and 18.11 for the five values of N ; takes $b = -a$ an exact double near $\bar{\omega}(N/2 + 1/4)$, which puts both horizontal edges a quarter of ω from the rows $\text{Im } \mu \approx \omega k$ where the near-axis eigenvalues lie; takes $K_e = \lfloor N/2 \rfloor + 16$, $n_c = 24$, $\eta = 2^{-20}$; uses Lemma 5.10 for the tail (the program calls this “route A”; an alternative “route B”, a direct cover of the bounded part described in `code/fourier/LEMMAS-stability.md`, is not used); chooses the exponents of S in floating point to make the tail bound small; computes V and Λ by LAPACK on the midpoint of the window block in the coordinates S , single-threaded; and decides every inequality of (C1) to (C5) in Arb. The count (C5) compares the exact doubles λ_j with the exact dyadic edges $-\delta$, R_0 , a , b . Floating point only chooses exact numbers, which decides whether the inequalities hold, never whether a passed inequality is true. Before any conclusion the program also checks that the residual of the trivial eigenvector P^* under the window block is small (about 10^{-20}), which would expose a dropped coefficient or a wrong sign convention.

Proof of Theorem B(c) and Theorem A(ii). For each $N \in \{1, 8, 16, 32, 64\}$ the records `data/fourier-stability-N*.json` show (C0) to (C5) passed, with Lemma 5.10 for the tail and the form (SC): for example at $N = 8$, $\rho_T = 3.578$, $\theta_c = 0.0787$, $\theta_T = 0.2817$, $\hat{b} = 0.856$, $q_C = 2.6 \times 10^{-14}$,

the smallest dist_j is 1.32×10^{-6} , and exactly one λ_j (at about $3 \times 10^{-16} - 4 \times 10^{-15}i$) lies in Ω ; Table 2 lists the other N . Theorem 5.17 gives the statements, with the multiplier bounds $e^{-\delta T_0}$ and $e^{-\delta T_0/N}$ rounded upward from Arb. For $N = 1$, $Y(T) = M_\tau$ and $Q = I$. $\square$

6 The cell in the time domain

The proof of Theorem A(i) uses CAPD [23]. Work in the scaled variables z of Section 2.1, and let $\mathcal{S} = \{z_V = s/\sigma_V\}$ (that is $V = s$; $s/\sigma_V = 4s$ is an exact double) and g the first-return map to $\mathcal{S}$, crossed upward. The frame file `code/candidates/cell_frameF.txt`, whose SHA-256 the record `data/cell-gks0.0275.json` stores, gives a point $\hat{x}$ with $\hat{x}_V = s/\sigma_V$ exactly (the verifier refuses the file otherwise), an invertible 17×17 matrix A_t , and a partition of the 17 free coordinates into 14 blocks of size 1 or 2 with radii ρ_b between 1.13×10^{-13} and 8.36×10^{-11} (listed in the record).

With $A = \begin{pmatrix} 1 & 0 \\ 0 & A_t \end{pmatrix}$ the map $y \mapsto \hat{x} + A(0, y)$, $y \in \mathbb{R}^{17}$, is an affine parametrization of $\mathcal{S}$, since the first component of $A(0, y)$ is 0. Let $\|y\| = \max_b \|y_b\|_2/\rho_b$ and

$$B = \{\hat{x} + A(0, y) : \|y\| \leq 1\}, \quad G(y) = A_t^{-1}(g(\hat{x} + A(0, y)) - \hat{x})_{\text{free}}.$$

$\hat{x}$ and A_t are proposed by the untrusted programs `code/proofs/orbit_newton.cpp` and `code/proofs/frame.py` (eigenvectors of the numerical Dg for the slow directions and an orthonormal Schur basis for the fast ones). The program `code/proofs/verify.cpp` encloses A_t^{-1} rigorously, $G(0)$ by a 128-bit MPFR interval run (order 30) and DG over the whole box $\prod_b[-\rho_b, \rho_b] \supset \{\|y\| \leq 1\}$ by a C^1 Lohner run in double intervals (order 20), and checks:

- $q = \max_b \sum_c \|M_{bc}\| \rho_c/\rho_b < 1$, where M encloses DG over the box (exact spectral bounds on diagonal 2×2 blocks, Frobenius bounds elsewhere);
- $\|G(0)_b\|/\rho_b + (\text{row sum of block } b) < 1$ for every block b ;
- every bound entering the decision is finite, and the section time of the center run lies inside the section time of the box run, so both resolve the same crossing; CAPD's crossing step checks that dV/dt does not vanish on the enclosure of the crossing point, so the crossing is transversal and g is C^1 on B .

Then G maps $\{\|y\| \leq 1\}$ into itself and is a q -contraction there in the norm $\|\cdot\|$ (mean value inequality on a convex set), so g has a unique fixed point x_c in B and every eigenvalue of $Dg(x_c)$ has modulus at most q . By Corollary 5.6 with $N = 1$ ($Q = I$, $\tau = T$), $\det(\lambda I - Y(T)) = (\lambda - 1) \det(\lambda I - Dg(x_c))$, so these eigenvalues are the 17 nontrivial Floquet multipliers. Local orbital asymptotic stability, with asymptotic phase and rate $e^{-\delta' t}$ for every δ' with $e^{-\delta' T} > q$, follows from the proof of Theorem 5.17(iii) with Σ replaced by $\mathcal{S}$ near x_c , P by g (whose return time is C^1 by transversality), and the ball B_c by B with the norm $\|\cdot\|$, in which g is already a q -contraction. The period is the return time: since $g(x_c) = x_c$, the first return time t_1 of x_c is a period, and since the orbit is back at $x_c \in \mathcal{S}$ at its minimal period, t_1 is the minimal period. The fixed point lies in the relative interior of B in $\mathcal{S}$, as the proof of Theorem 5.17(iii) needs: by the second check, G maps $\{\|y\| \leq 1\}$ into $\{\|y\| \leq 0.99865\}$, and x_c corresponds to a fixed point of G . The record gives $q \leq 0.99864150686405151$ (stored as the double `0x1.fff4df088d18aap-1`), $\max_b (\|G(0)_b\|/\rho_b + \text{row sum}) \leq 0.99864258833469344$, and the return time `[53.5855190480139, 53.585519630722438]` ms; the run took 732 s.

Lemma 6.1 (the two cell certificates enclose the same orbit). *Let ϕ_* be the profile of Theorem A(ii) and $p = \Sigma^{-1}\phi_*(0)$. Then $p \in B$, and the orbit $\{\Sigma^{-1}\phi_*(\theta)\}$ of Theorem A(ii), in scaled variables, is the orbit of the fixed point x_c of Theorem A(i).*

Proof. $p_V = s/\sigma_V$ by the phase condition, so $p \in \mathcal{S}$. For each free component k , $|p_k - \bar{p}_k| \leq \sum_m |a_{*,k,m} - \bar{a}_{k,m}| \leq \|a_{*,k} - \bar{a}_k\|_\nu \leq r_{\text{ex}}$, where $\bar{p} = \sum_{|m| \leq K} \bar{a}_m$ is an exact rational vector (the program checks that $\text{Im } \bar{a}_0 = 0$, so $\bar{p} = \bar{a}_0 + 2 \text{Re } \sum_{m=1}^K \bar{a}_m$). The program `code/fourier/link_cell.py` reads $\hat{x}$, A_t and the radii as the doubles the verifier reads, inverts A_t exactly in rational arithmetic, computes $y^c = A_t^{-1}(\bar{p} - \hat{x})_{\text{free}}$ exactly and the bound $|y_k - y_k^c| \leq r_{\text{ex}} \sum_j |(A_t^{-1})_{kj}|$ for $y = A_t^{-1}(p - \hat{x})_{\text{free}}$, and checks, with upper bounds of square roots in integer arithmetic, $\|y_b^c\|_2 + \|(y - y^c)_b\|_2 < \rho_b$ for every block. All 14 checks pass, the largest ratio being below 0.14 (`data/link_cell.txt`); a control with every radius divided by 10 fails. So $p \in B$. Since $g(B) \subset B$ and g is a contraction on B , $g^k(p) \in B$ and $g^k(p) \rightarrow x_c$. Each $g^k(p)$ is a point of the flow line through p of the CAPD field; by the assumptions of Theorem A(i) and (ii) taken together, the CAPD and Arb fields are the same function, so that flow line is the closed set $\{\Sigma^{-1}\phi_*(\theta)\}$. So x_c lies on it, and the orbit of x_c is that set. Neither the crossing direction of p nor $x_c = p$ is used: CAPD defines g on all of B , and only that $g^k(p)$ lies on the flow line of p is needed. $\square$

The domain and the CAPD patch. CAPD's interval wrapper throws when a divisor contains 0, a logarithm argument is not positive or a square-root argument is negative, and the run then fails; the multiprecision logarithm does not check its argument, so every $\log a$ is computed as $2 \log \sqrt{a}$ (the same function for $a > 0$), and the multiprecision square root refuses a negative argument. The source's $(1 - u)$ is computed as $1/(1 + e^{5(V+40)})$, the same function, because near $V = 0$, u is within 10^{-80} of 1. In `PoincareMap::crossSectionInOneStep` of CAPD 6.1.0 the crossing-point bound is trimmed with the endpoints of the previous Newton window while monotonicity was checked on the current one, which is unsound if the loop stops at its iteration cap without converging. The run used a header-only local patch that takes the endpoints of the current window (`code/proofs/capd-6.1.0-genchase.patch`); the record stores the SHA-256 of the patched header. An earlier unpatched run gave the same bounds.

7 Numerical observations

Nothing in this section is proved, and none of it is used in a proof.

- *Leading exponents.* The floating-point eigenvalues of the window blocks closest to the imaginary axis, apart from 0, have real parts -4.71382×10^{-5} ($N = 1$), -6.32096×10^{-6} ($N = 8$), -8.57417×10^{-6} (16), -9.18618×10^{-6} (32) and -9.34235×10^{-6} per ms (64), with imaginary parts near 0 ($N = 1$) and near $\pm\omega$ (the rings). The certified $\delta = 5 \times 10^{-6}$ for the rings is below all of them.
- *Sharpness at $N = 8$.* In unrecorded runs of `code/fourier/stability.py`, the certificate passes at $N = 8$ with $\delta = 6.32095 \times 10^{-6}$ (near-axis (SC) ratio 0.398) and fails at $\delta = 6.321 \times 10^{-6}$ with a count of 3 in (C5). So the certificate is sharp to about 10^{-6} relative against the floating-point exponent; we do not state the larger δ as a theorem.
- *Negative controls.* In the same runs the certificate failed, at the step that is supposed to fail, for $\delta = 7 \times 10^{-6}$ at $N = 8$ ((C5)), for reversed coupling $c \rightarrow -c$ ((C5), count 9), for a window that is too small (Lemma 5.10), for $S = I$ (Lemma 5.10 or $\theta_T < 1$ fails), for a coefficient $A_{\pm 1}$ dropped from the proof data only ((SC) fails in 127 columns), and for floating-point data that place the leading pair at -7.5×10^{-6} while the true data do not ((SC) fails).
- *Hopf point.* An ordinary finite-difference computation (`code/numerics/hopf_and_orbit.py`, `data/numerics-hopf-orbit.json`) reproduces Erhardt's Hopf point to 5×10^{-7} relative

(0.0279078439 against 0.0279078589) and the Hopf frequency to 3×10^{-7} relative. In that computation the voltage of the cell orbit stays between about 0.14 and 0.26 mV: a small depolarized oscillation, not an action potential.

- *Ring periods.* The certified periods increase with N by 2.45×10^{-3} , 9.8×10^{-5} , 2.5×10^{-5} and 6.3×10^{-6} ms from $N = 1$ to 8, 8 to 16, 16 to 32 and 32 to 64. The ratios of successive differences, about 3.9 and 4.0 for the last two, look like second-order convergence in $1/N$. Theorem C(d) proves $T_N \rightarrow T_\infty$ but no rate; the extrapolation $T_{64} + (T_{64} - T_{32})/3 \approx 53.5881025$ ms lies inside its interval for T_∞ , which is an observation, not a proof of second-order convergence.
- *Controls for Theorem C.* Recorded in `data/fourier-existence-alln.json` (from `code/fourier/data/alln/controls.jsonl`) and not used in the proof: on the piece $[0, 1/4096]$, omitting the terms of the parameter width ($\delta_P Y_0^g$ and $\delta_P B_1^g$) lowers Y_0 to 3.4×10^{-11} , below the floating-point norms 7.25×10^{-7} of $A_{\text{fin}} F(\bar{x}; \varepsilon)$ at the two ends of the piece, so those terms are needed; single pieces $[0, w]$ fail for $w = 1/16, 1/8, 1/4$ and 1 (with Z_1 between 1.27 and 16.1), and close for $w = 1/1024, 1/256$ and $1/64$ ($Z_1 = 0.30$ for $[0, 1/64]$); those closed pieces are not part of the record's cover.

Remark 7.1 (phase reduction; numerical, not part of the proofs). We compared the rotating 1-waves with the classical phase reduction of weakly coupled oscillators [3, 20], in floating point (`code/numerics/phase_reduction.py`, `data/numerics-phase-reduction.json`); nothing in this remark is proved or used in a proof. Let $Z(\theta)$ be the infinitesimal phase response of the cell orbit $\phi(\omega t)$, the gradient of its asymptotic phase (in radians) at $\phi(\theta)$, normalized by $Z \cdot \phi' = 1$ and computed spectrally from the adjoint equation $\omega Z' = -Df(\phi)^\top Z$; at four phases, a finite-difference monodromy computation over one period agrees with it to within 3×10^{-5} relative in the V component. Let

$$H(\psi) = \frac{1}{2\pi} \int_0^{2\pi} Z_V(\theta) (\phi_V(\theta + \psi) - \phi_V(\theta)) d\theta.$$

The averaged phase model of the ring, $\dot{\theta}_j = \omega + c [H(\theta_{j-1} - \theta_j) + H(\theta_{j+1} - \theta_j)]$, which has the form studied by Ermentrout [15], has the splay state $\theta_j = \Omega t + 2\pi j/N$ for every N , with $\Omega = \omega + c [H(q) + H(-q)]$, $q = 2\pi/N$. Its ring mode k has the exponent $\lambda_k = c [H'(q)(e^{ikq} - 1) + H'(-q)(e^{-ikq} - 1)]$, so the splay state is stable if and only if $H'(q) + H'(-q) > 0$. For $N = 8, 16, 32, 64$ the computed $H'(q)$ and $H'(-q)$ are both positive, which is Ermentrout's sufficient condition for stability [15, Theorem 3.1 and Lemma 3.2].

- *Predictions.* For $N = 8, 16, 32, 64$ the predicted period shifts $T_N - T_1$ fall short of the certified ones (2.452×10^{-3} to 2.581×10^{-3} ms, Table 1) by 0.77 to 0.82 per cent. The predicted leading nontrivial exponent ($k = 1$) differs from the floating-point leading exponent of this section by 3.2 to 3.5 per cent in its real part and by 2.2 to 2.7 per cent in its imaginary part, imaginary parts being compared modulo ω , as Floquet exponents are defined only modulo $i\omega$.
- *Weak coupling.* With the coupling scaled by a factor $\sigma \in (0, 1]$ and the wave recomputed by Newton's method, both relative discrepancies of the leading exponent, and that of the period shift, decrease in proportion to σ (for the real part at $N = 8$: 4.9×10^{-4} at $\sigma = 1/64$ and 3.1×10^{-2} at $\sigma = 1$). So the leading ring modes are in the weak-coupling regime. The relevant parameter $4c \sin^2(\pi/N)/\omega$ lies between 5.0×10^{-3} and 5.3×10^{-3} for the four N .
- *Where the phase model fails.* It is not uniformly valid. For $N \geq 16$ its decay rates for the short-wavelength ring modes ($k \geq 3$) exceed the slowest transverse decay rate of the cell,

4.71×10^{-5} per ms, while the floating-point leading exponents of those modes stay between -5.74×10^{-5} and -4.75×10^{-5} per ms; at $N = 64$, $k = 32$, the phase model predicts -4.0×10^{-3} per ms.

The phase reduction thus accounts for the size of the leading exponents, and so for the scale of the certified $\delta = 5 \times 10^{-6}$ per ms, but it cannot replace the Hill-operator certificate for the whole spectrum.

8 Reproducibility and trust base

The programs are in `code/`, in the layout of their canonical location `research/cardiac-cycle-certificates/` of the GENChase repository, whose records hash them; the records in `data/` are byte-identical copies, and `code/fourier/check_records.py` rechecks all 90 stored hashes (sources, centers and the Stage E records read by Stage S) in a few seconds. `code/run_all.sh` stages the programs and records in a scratch folder, runs that check and the link of Lemma 6.1 (`code/fourier/link_cell.py`, standard library only), and reruns Stage E and Stage S for chosen N . For Theorem C it also checks the 15 hashes stored in `data/fourier-existence-alln.json` (its nine sources, among them `code/fourier/alln.py` and `code/fourier/branch.py`, its run log `code/fourier/data/alln/pieces.jsonl`, which holds the exact inputs of every piece, and the Stage E records it identifies), and with the argument `alln` it re-derives in Arb, from the stored exact data, the piece order, every gluing inequality (10) and the inclusions (11) (`alln.py-collect`), and compares them with the record. Re-proving pieces from their stored inputs is done by `code/fourier/test_alln.py`.

Fourier route. Trusted: Arb and FLINT through python-flint 0.9.0 (FLINT 3.6.0; the wheel’s SHA-256 is pinned and checked by `code/fourier/test_arbmodel.py`), CPython, the programs `code/fourier/arbmodel.py`, the generated `code/fourier/tp06_18d_arb.py`, `fourier_eval.py`, `existence.py`, `stability.py`, for Theorem C also `alln.py` and the functions of `branch.py` it calls (the assembly of the bounds, the Hessian cover with its second-order dual numbers, the distance of centers and the check of the piece order), and the reference translation `code/model/tp06_18d.py` with the scales. The generator rewrites every decimal literal of the reference translation as an exact rational by Python’s tokenizer, and a freshness check compares the generated file byte for byte. numpy and LAPACK only propose centers, eigenvectors and weights. The tests compare the Arb field with an mpmath evaluation and with the CAPD field at random points, test the strip cover, the aliasing bound and the centered form against closed forms, and contain the negative controls of Section 7, independent floating-point estimates that every certified bound must dominate, and perturbation controls within the uniqueness ball.

CAPD route. Trusted: CAPD 6.1.0 at commit 03dc5628203334b214bb7d9fd63788a175521005 with the local patch above, filib and MPFR, g++ 13.3 with `-frounding-math`, the processor’s directed rounding, and `code/proofs/verify.cpp` with `code/model/tp06_capd.hpp` and `setup.hpp`, in which every non-integer decimal enters as an interval enclosing that exact decimal. The driver `code/proofs/certify.py` checks that the frame file’s header names the intended claim and records the hashes of the frame, the binary and the sources.

What is certified, and reruns. The theorems rest on the stored records in `data/`: the Stage S records hash the stored Stage E records they read, so the stored r_{ex} is the one consumed. On

2026-10-01 the existence and stability programs for $N = 1$ and $N = 8$ were rerun from the copies in `code/` with python-flint 0.9.0 (`notes/rerun-2026-10-01.md`): the period and frequency enclosures, r_{un} and every stability bound of Table 2 agree exactly with the records, while the binary values of Y_0 , Z_1 , Z_2 and r_{ex} differ from the records in late digits (for Y_0 beyond the 25 printed digits, for the others in about the thirteenth significant digit). The likely cause is that `existence.py`, unlike `stability.py`, does not pin the number of BLAS threads, so the untrusted floating-point inverse A_{fin} need not be bit-reproducible; any exact A_{fin} is admissible, and only the size of the bounds depends on it. The rings $N = 16, 32, 64$ and the CAPD certificate were not rerun from these copies. The pieces of Theorem C were not re-proved from these copies; on 2026-10-02 the collection step (`sh code/run_all.sh alln`) was rerun from them and reproduced the piece table, the gluing and the inclusions of the record exactly. `alln.py`, unlike `existence.py`, pins the number of BLAS threads to one.

The run times quoted are those of the records: about two minutes for each existence proof, and 26, 48, 80, 184 and 479 s for the stability proofs at $N = 1, 8, 16, 32, 64$ (peak memory 3.6 GB at $N = 64$); the 73 pieces of Theorem C took 3129 s of computing time in all, between 28 and 63 s each.

9 Limitations and what has been checked

- *Scope.* The theorems are about one modified cell model at one parameter value, $G_{Ks} = 0.0275$, and about rings with one coupling law: existence, local uniqueness and the minimal period for every $N \geq 8$ and for the cable (7) on a ring of unit length (Theorem C), and stability only for $N = 8, 16, 32, 64$ (Theorem B). Nothing is claimed for $N < 8$, for the stability of the waves with other N or of the cable wave, for stability uniformly in N , for a rate of convergence as $N \rightarrow \infty$, for the published 19-state TP06 cell, for action potentials or reentry, or for tissue in two or three dimensions. For $N \geq 8$ other than 8, 16, 32, 64 the period is enclosed only to about 1.2×10^{-5} ms. The orbits are small depolarized oscillations (Section 7).
- *The Hopf branch.* That the cell orbit lies on the branch of periodic orbits born at Erhardt’s Hopf point, and that the ring waves continue it, is supported by numerics only. Theorem C connects the ring waves with each other and with the cable wave through ε , at the fixed value $G_{Ks} = 0.0275$; it says nothing about a continuation in G_{Ks} .
- *Future work: an interval in G_{Ks} .* A certified branch of these orbits on an interval of G_{Ks} toward the Hopf point is future work; no result of it is used or claimed here.
- *Translations of the model.* Theorem A(i) assumes that the CAPD field (`code/model/tp06_capd.hpp`) evaluates Erhardt’s `fun_eval` with exact decimals, and (ii), Theorem B and Theorem C assume the same of the Arb field (`code/fourier/tp06_18d_arb.py`); Lemma 6.1 uses both assumptions together. The tests compare the two programs’ enclosures at 104 points, which is evidence, not a proof that they define the same function.
- *Operators with compact resolvent.* The three facts of Section 5.1 are proved in Appendix A from elementary facts (the Neumann series, the Hahn–Banach theorem, the Cauchy–Goursat and identity theorems, Jordan decomposition in finite dimension); Kato’s book is cited as the standard reference only, and no step depends on it. The appendix, added on 2026-10-02, had an adversarial reading on the same day by a separate AI agent session within the project, covering Appendix A and its uses in Section 5 (`review/appendixA-reading-2026-10-02.md`); it found no error, two gaps in how Section 5 hands its operators to the appendix (the space of

Lemma 5.9; the domains of $\hat{D}$ and D_T and the boundedness of $\hat{E}$) and exposition items, all addressed in this version. This is not an outside review.

- *How far the background sources were read.* None of them is a source of a proof step; only Erhardt’s model source and [12] (read in full) enter the setting. Read in part: Gameiro and Lessard [17] (Sections 1, 7 and 8 of the authors’ version), Bayer and Leine [4] (Sections 1 to 4, 6 and 7 of arXiv:2503.21318v2), Ermentrout [15] (pp. 1674–1677), Johnson and Zumbrun [22] (first page), Czechowski and Zgliczyński [9] (Theorems 1.1 to 1.3). Known from abstracts: van den Berg, Lessard and Mischaikow [33], Paullet and Ermentrout [29], Church and Lessard [7], Arioli and Koch [2], Kapela and Zgliczyński [24], Kuehn and Queirolo [26], ten Tusscher and Panfilov [32], Church et al. [6], Castelli and Lessard [5], Rucklidge and Silber [31], de Wolff [10], the papers of Erhardt and Solem [13, 14] and of Kügler et al. [27, 28]. Known from its title and Czechowski and Zgliczyński’s account: Arioli and Koch [1]. Known from metadata or a table of contents: Golubitsky, Stewart and Schaeffer [19] (table of contents), Golubitsky and Stewart [18], Figueras et al. [16], Courtemanche et al. [8], Pecora and Carroll [30], Ashwin and Swift [3], Hoppensteadt and Izhikevich [20]. Known from excerpts of its abstract: Di Marco et al. [11]. The notes in `research/cardiac-cycle-certificates/notes/` of the GENChase repository record each reading.
- *Searches.* The prior-article searches of 2026-09-30 and 2026-10-01 are recorded in `research/cardiac-cycle-certificates/notes/` of the GENChase repository. The papers citing Erhardt [12] were not checked for a continuation of the first Hopf branch. The searches did not cover traveling waves of continuum cables of detailed ionic models, so no novelty is claimed for Theorem C(c).
- *Checks.* Each program and each stability lemma had an adversarial reading by a separate AI agent session within the project, followed by a second reading of the fixes, which found nothing unsound; the CAPD verifier had an adversarial review within the project, which led to the patch above; and the manuscript had two such readings, whose findings are addressed in this version. The program of Theorem C, `code/fourier/alln.py`, with its tests, record and the parts of `branch.py` it calls, had an adversarial reading by a separate AI agent session within the project (`review/alln-existence-review-2026-10-02.md`); it found nothing unsound and no gap affecting the theorem, re-proved two pieces bit for bit, and reported four weak tests and four minor items, which the program’s author fixed without changing any bound (`review/alln-existence-fixcheck-2026-10-02.md`, not an independent reading); the record keeps the status its program wrote. The integration of Theorem C and Remark 7.1 into this manuscript had a further in-project reading (`review/alln-integration-reading-2026-10-02.md`). The reports are in `review/`. None of these is an outside review. Two items of the program readings remain open: a few negligible terms (for example the strip tail of S_J beyond K') cannot be detected by any test, and the source hashes are taken after the modules are imported, a small window.

10 Discussion

The method separates the dimension of the ring from the cost of the proof: existence costs the same for every N , and stability grows with the window $K_e \approx N/2 + 16$ that must contain the N near-axis copies of the spectrum. The Hill-sector theorem turns a $18N$ -dimensional Floquet problem into a question about one operator on $\ell^1(\mathbb{Z}; \mathbb{C}^{18})$, and the Riesz homotopy with the Schur-complement

small gain answers it with a count, which is what a stability proof needs and an eigenpair enclosure cannot give. The margins are thin: at $N = 8$ the leading nontrivial exponent is about -6.32×10^{-6} per ms, so the multipliers are within about 3.4×10^{-4} of the unit circle per period; the certificate still resolves them because the window is computed at 128 bits and the near-axis columns of (SC) have ratios below 10^{-6} . The phase reduction of Remark 7.1 accounts for the size of these leading exponents, but not for the rest of the spectrum. For existence the dependence on N is removed altogether: since the damping is an entire function of $\varepsilon = 1/N^2$, all rings with $N \geq 8$ and the cable are one family, and 73 pieces in ε prove all of them at a cost of under an hour of computing time (Theorem C). Stability does not yet follow this route: Stage S certifies the spectrum of one Hill operator, and a version uniform over a piece in ε has not been attempted. Natural next steps are stability uniform in N and for the cable, and an interval of G_{Ks} (in progress).

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Use of AI. This work was prepared with AI assistance. The author takes full responsibility for its content.

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A Operators with compact resolvent on ℓ^1 spaces

This appendix proves the facts about operators with compact resolvent that Section 5 uses, at the generality needed there. They are standard; Kato’s book [25, Ch. III, Sect. 6] is the usual reference, and nothing below depends on it. The only result of the body that is used is Lemma 5.8 (two bounded projections at distance less than 1 have the same finite rank), whose three-line proof does not depend on anything else. Besides standard facts of linear algebra (determinants and Cramer’s rule, boundedness of coordinate functionals on a finite-dimensional normed space), of metric spaces (a relatively compact set has a finite ϵ -net) and of the Riemann integral (a uniformly continuous function on a compact interval, with values in a Banach space, has a Riemann integral, the norm limit of its Riemann sums), and the Neumann series, we use only the following: the Hahn–Banach theorem (a nonzero element of a normed space is not annihilated by every bounded linear functional); the Cauchy–Goursat theorem for a function holomorphic on an open set containing a closed rectangle; the identity theorem (on a connected open set, the zeros of a holomorphic function that is not identically zero have no accumulation point in that set); for the positively oriented boundary Γ of a bounded open rectangle Ω and $\mu \notin \Gamma$, $\frac{1}{2\pi i} \oint_{\Gamma} (\lambda - \mu)^{-1} d\lambda$ is 1 for $\mu \in \Omega$ and 0 for $\mu \notin \bar{\Omega}$, and $\oint_{\Gamma} (\lambda - \mu)^{-k} d\lambda = 0$ for $k \geq 2$ (a primitive exists); Fubini’s theorem for continuous functions on a product of two compact intervals; and, for a linear map T of a finite-dimensional complex space M , the decomposition $M = \bigoplus_{\mu \in \text{spec } T} \ker(T - \mu)^{\dim M}$ and the boundedness of T in any norm.

Setting. X is a complex Banach space and $B(X)$ the bounded operators on it; an element of $B(X)$ is invertible if it is bijective with a bounded inverse. For a linear operator H with domain $D(H) \subset X$, the resolvent set $\rho(H)$ is the set of $\lambda \in \mathbb{C}$ for which $\lambda - H : D(H) \rightarrow X$ is bijective with a bounded inverse $R(\lambda) = (\lambda - H)^{-1}$, and $\text{spec } H = \mathbb{C} \setminus \rho(H)$. If $\rho(H) \neq \emptyset$, H is closed: if $x_n \rightarrow x$ and $Hx_n \rightarrow y$, then $x_n = R(\lambda)(\lambda - H)x_n \rightarrow R(\lambda)(\lambda x - y)$, so $x = R(\lambda)(\lambda x - y) \in D(H)$ and $Hx = y$. H has compact resolvent if $R(\lambda_0)$ is compact for some $\lambda_0 \in \rho(H)$. We write $\ker(H - \mu)^k$ for the set of $x \in D(H)$ with $(H - \mu)^j x \in D(H)$ for $j < k$ and $(H - \mu)^k x = 0$, $G_\mu(H) = \bigcup_k \ker(H - \mu)^k$, $m(\mu; H) = \dim G_\mu(H) \in \{0, 1, \dots, \infty\}$ and $n(H, \Omega) = \sum_{\mu \in \Omega} m(\mu; H)$, as in Section 5.1. A rectangle is a bounded open set $\Omega = (x_0, x_1) \times (y_0, y_1) \subset \mathbb{C}$, and $\Gamma = \partial\Omega$, positively oriented, of length $|\Gamma|$. An operator is approximable if it is a norm limit of finite-rank operators in $B(X)$. Approximable operators form a norm-closed linear subspace of $B(X)$, and AKB is approximable whenever K is and $A, B \in B(X)$ ($\|AKB - AFB\| \leq \|A\|\|K - F\|\|B\|$ and AFB has finite rank).

The spaces of Section 5 are $\ell_w^1(\mathcal{J}) = \{x : \|x\| = \sum_{i \in \mathcal{J}} w_i |x_i| < \infty\}$ with $\mathcal{J}$ countable and weights $w_i > 0$: $\mathcal{X} = \ell^1(\mathbb{Z}; \mathbb{C}^{18})$ with $\mathcal{J} = \mathbb{Z} \times \{1, \dots, 18\}$, and the space of the Scal-coordinates of Section 5.4, with $\mathcal{J}$ the window coordinates together with $\text{Tl} \times \{1, \dots, 18\}$, all weights 1.

Lemma A.1. *On $X = \ell_w^1(\mathcal{J})$ every compact operator is approximable.*

Proof. For finite $\mathcal{J}$ there is nothing to prove. Otherwise enumerate $\mathcal{J} = \{i_1, i_2, \dots\}$ and let Π_n keep the coordinates $i_1, \dots, i_n$ and set the others to 0. Then Π_n has finite rank, $\|I - \Pi_n\| \leq 1$, and $\|x - \Pi_n x\| = \sum_{k > n} w_{i_k} |x_{i_k}| \rightarrow 0$ for every x . Let K be compact and $\epsilon > 0$. The image of the unit ball under K is relatively compact, so it has a finite ϵ -net $y_1, \dots, y_r$; choose n with $\|y_l - \Pi_n y_l\| < \epsilon$ for every l . For $\|x\| \leq 1$ pick l with $\|Kx - y_l\| < \epsilon$; then $\|(I - \Pi_n)Kx\| \leq \|(I - \Pi_n)(Kx - y_l)\| + \|(I - \Pi_n)y_l\| < 2\epsilon$. So $\|K - \Pi_n K\| \leq 2\epsilon$, and $\Pi_n K$ has finite rank. $\square$

Lemma A.2 (the resolvent). *Let $\rho(H) \neq \emptyset$.*

- (a) *For $\lambda, \mu \in \rho(H)$, $R(\lambda) - R(\mu) = (\mu - \lambda)R(\lambda)R(\mu)$, and $R(\lambda)R(\mu) = R(\mu)R(\lambda)$.*
- (b) *$R(\lambda)$ maps X onto $D(H)$, $HR(\lambda) = \lambda R(\lambda) - I \in B(X)$, and $R(\lambda)Hx = HR(\lambda)x$ for $x \in D(H)$.*
- (c) *If $\lambda_0 \in \rho(H)$ and $|\lambda - \lambda_0|\|R(\lambda_0)\| < 1$, then $\lambda \in \rho(H)$ and $R(\lambda) = \sum_{k \geq 0} (\lambda_0 - \lambda)^k R(\lambda_0)^{k+1}$. So $\rho(H)$ is open, R is norm continuous on $\rho(H)$, and R is holomorphic there: $(R(\lambda) - R(\mu))/(\lambda - \mu) \rightarrow -R(\mu)^2$ in norm as $\lambda \rightarrow \mu$.*
- (d) *If $R(\lambda_0)$ is approximable for one $\lambda_0 \in \rho(H)$, then $R(\lambda)$ is approximable for every $\lambda \in \rho(H)$.*
- (e) *If $\mathcal{E} \in B(X)$, $\lambda \in \rho(H)$ and $I - \mathcal{E}R(\lambda)$ is invertible, then $\lambda \in \rho(H + \mathcal{E})$, with $D(H + \mathcal{E}) = D(H)$ and $(\lambda - H - \mathcal{E})^{-1} = R(\lambda)(I - \mathcal{E}R(\lambda))^{-1}$. If $\lambda \in \rho(H) \cap \rho(H + \mathcal{E})$, then $(\lambda - H - \mathcal{E})^{-1} - R(\lambda) = (\lambda - H - \mathcal{E})^{-1}\mathcal{E}R(\lambda)$.*

Proof. (a) Since $R(\mu)$ maps into $D(H)$, $R(\lambda) = R(\lambda)(\mu - H)R(\mu)$ and $R(\mu) = R(\lambda)(\lambda - H)R(\mu)$; subtract. Exchanging λ and μ and comparing gives commutativity for $\lambda \neq \mu$. (b) $R(\lambda)X \subset D(H)$ by definition, and every $x \in D(H)$ is $R(\lambda)(\lambda - H)x$, so $R(\lambda)X = D(H)$. $(\lambda - H)R(\lambda) = I$ gives the formula for $HR(\lambda)$; for $x \in D(H)$, $R(\lambda)(\lambda - H)x = x$ gives $R(\lambda)Hx = \lambda R(\lambda)x - x = HR(\lambda)x$. (c) Let $T = (\lambda_0 - \lambda)R(\lambda_0)$, so $\|T\| < 1$. For $x \in D(H)$, $(\lambda - H)x = (\lambda_0 - H)x - (\lambda_0 - \lambda)x = (I - T)(\lambda_0 - H)x$. Since $I - T$ is invertible by the Neumann series, $\lambda - H$ is a bijection $D(H) \rightarrow X$ with the bounded inverse $R(\lambda_0)(I - T)^{-1} = \sum_{k \geq 0} (\lambda_0 - \lambda)^k R(\lambda_0)^{k+1}$, and $\|R(\lambda) - R(\lambda_0)\| \leq \sum_{k \geq 1} |\lambda - \lambda_0|^k \|R(\lambda_0)\|^{k+1} \rightarrow 0$ as $\lambda \rightarrow \lambda_0$. The derivative follows from (a) and continuity. (d) By (a), $R(\lambda) = R(\lambda_0) + (\lambda_0 - \lambda)R(\lambda)R(\lambda_0)$. (e) For $x \in D(H)$, $(\lambda - H - \mathcal{E})x = (I - \mathcal{E}R(\lambda))(\lambda - H)x$, a composition of bijections with bounded inverses. The last identity is $(\lambda - H - \mathcal{E})^{-1}[(\lambda - H) - (\lambda - H - \mathcal{E})]R(\lambda)$. $\square$

For a continuous $F : \Gamma \rightarrow B(X)$, $\oint_{\Gamma} F(\lambda) d\lambda$ is the sum over the four edges of the norm limits of Riemann sums, which exist by uniform continuity exactly as for scalar functions.

Lemma A.3 (contour integrals). *Let Ω be a rectangle and $F : \Gamma \rightarrow B(X)$ continuous.*

- (a) $\|\oint_{\Gamma} F\| \leq |\Gamma| \max_{\Gamma} \|F\|$; $A(\oint_{\Gamma} F)B = \oint_{\Gamma} AFB$ for $A, B \in B(X)$; $(\oint_{\Gamma} F)x = \oint_{\Gamma} F(\lambda)x d\lambda$; $\ell(\oint_{\Gamma} F) = \oint_{\Gamma} \ell \circ F$ for every bounded linear functional ℓ on $B(X)$.
- (b) If every $F(\lambda)$ is approximable, so is $\oint_{\Gamma} F$.
- (c) If H is closed, $F(\lambda)X \subset D(H)$ for every λ , and $\lambda \mapsto HF(\lambda)$ is continuous into $B(X)$, then $(\oint_{\Gamma} F)X \subset D(H)$ and $H \oint_{\Gamma} F = \oint_{\Gamma} HF$.
- (d) If F extends to a holomorphic function on an open set containing $\overline{\Omega}$, then $\oint_{\Gamma} F = 0$. If Ω' is Ω with each side moved outward by $\epsilon > 0$, and F is holomorphic on an open set containing $\overline{\Omega'} \setminus \Omega$, then $\oint_{\Gamma'} F = \oint_{\Gamma} F$, $\Gamma' = \partial\Omega'$.
- (e) If Γ' is the boundary of another rectangle and $G : \Gamma \times \Gamma' \rightarrow B(X)$ is continuous, then $\oint_{\Gamma'} \oint_{\Gamma} G(\lambda, \lambda') d\lambda' d\lambda = \oint_{\Gamma'} \oint_{\Gamma} G(\lambda, \lambda') d\lambda d\lambda'$.

Proof. (a) and (b) hold for Riemann sums and pass to the norm limit. (c) The Riemann sums S_k converge to $\oint F$ in norm, $S_k x \in D(H)$, and $HS_k x$ are the Riemann sums of $\lambda \mapsto HF(\lambda)x$, which converge to $(\oint HF)x$; H is closed. (d) For every bounded linear functional ℓ on $B(X)$, $\ell \circ F$ is holomorphic on that open set, so $\ell(\oint_{\Gamma} F) = \oint_{\Gamma} \ell \circ F = 0$ by the Cauchy–Goursat theorem, and $\oint_{\Gamma} F = 0$ by the Hahn–Banach theorem. For the second statement, $\overline{\Omega'} \setminus \Omega$ is the union of four closed rectangles: the two of width ϵ along the vertical sides of Ω' , of the full height of Ω' , and the two of height ϵ above and below Ω , of the width of Ω . The integrals over their positively oriented boundaries are 0 by the first statement, and they add up to $\oint_{\Gamma'} F - \oint_{\Gamma} F$, because every edge shared by two of them is traversed once in each direction, the outer edges make up Γ' , and the edges on Γ are traversed opposite to the orientation of Γ . (e) The inner integrals are continuous in the outer variable by uniform continuity of G . Apply ℓ as in (d); on each pair of edges the scalar statement is Fubini’s theorem for a continuous function on a product of two compact intervals; conclude by the Hahn–Banach theorem. $\square$

Proposition A.4 (Riesz projection). *Let $\rho(H) \neq \emptyset$, Ω a rectangle with $\Gamma \subset \rho(H)$, and $P = \frac{1}{2\pi i} \oint_{\Gamma} R(\lambda) d\lambda$.*

- (a) $P^2 = P$; $PR(\mu) = R(\mu)P$ for $\mu \in \rho(H)$; $PX \subset D(H)$, $HP \in B(X)$, and $PHx = HPx$ for $x \in D(H)$. So $M = PX$ is a closed subspace and $H(M) \subset M$.
- (b) If $x \in \ker(H - \mu)^k$ for some $k \geq 1$, then $Px = x$ if $\mu \in \Omega$ and $Px = 0$ if $\mu \notin \overline{\Omega}$.
- (c) For $\zeta \in \Omega$, $Q(\zeta) = \frac{1}{2\pi i} \oint_{\Gamma} (\lambda - \zeta)^{-1} R(\lambda) d\lambda$ maps X into $D(H)$, $(\zeta - H)Q(\zeta) = I - P$, and $Q(\zeta)(\zeta - H)x = (I - P)x$ for $x \in D(H)$.
- (d) If $\dim M < \infty$, then $\text{spec } H \cap \Omega$ is the spectrum of the linear map $H|_M$ of M , a finite set of eigenvalues of H , $M = \bigoplus_{\mu \in \text{spec } H \cap \Omega} G_{\mu}(H)$, and $\text{rank } P = n(H, \Omega)$.
- (e) If $R(\lambda)$ is approximable for every $\lambda \in \Gamma$, then $\dim M < \infty$.

Proof. (a) Γ is compact and $\text{spec } H$ is closed, so if $\text{spec } H \neq \emptyset$ they are at a distance $d_0 > 0$ (otherwise let $d_0 = 1$). Let Ω' be Ω with each side moved outward by $\epsilon = d_0/2$. Every point of $\overline{\Omega'} \setminus \Omega$ lies within $\sqrt{2}\epsilon < d_0$ of Γ , so $\overline{\Omega'} \setminus \Omega \subset \rho(H)$, an open set on which R is holomorphic (Lemma A.2(c));

by Lemma A.3(d), $P = \frac{1}{2\pi i} \oint_{\Gamma'} R(\lambda') d\lambda'$. By Lemma A.3(a), Lemma A.2(a) and Lemma A.3(e), since $\Gamma \cap \Gamma' = \emptyset$,

$$\begin{aligned} P^2 &= \frac{1}{(2\pi i)^2} \oint_{\Gamma} \oint_{\Gamma'} R(\lambda) R(\lambda') d\lambda' d\lambda = \frac{1}{(2\pi i)^2} \oint_{\Gamma} \oint_{\Gamma'} \frac{R(\lambda) - R(\lambda')}{\lambda' - \lambda} d\lambda' d\lambda \\ &= \frac{1}{2\pi i} \oint_{\Gamma} R(\lambda) \left[\frac{1}{2\pi i} \oint_{\Gamma'} \frac{d\lambda'}{\lambda' - \lambda} \right] d\lambda - \frac{1}{2\pi i} \oint_{\Gamma'} R(\lambda') \left[\frac{1}{2\pi i} \oint_{\Gamma} \frac{d\lambda}{\lambda' - \lambda} \right] d\lambda'. \end{aligned}$$

For $\lambda \in \Gamma \subset \Omega'$ the first bracket is 1; for $\lambda' \in \Gamma'$, which lies outside $\bar{\Omega}$, the second is 0. So $P^2 = P$. $PR(\mu) = R(\mu)P$ by Lemmas A.2(a) and A.3(a). By Lemmas A.2(b) and A.3(c), $PX \subset D(H)$ and $HP = \frac{1}{2\pi i} \oint_{\Gamma} (\lambda R(\lambda) - I) d\lambda = \frac{1}{2\pi i} \oint_{\Gamma} \lambda R(\lambda) d\lambda$, since $\oint_{\Gamma} d\lambda = 0$. For $x \in D(H)$, $R(\lambda)Hx = \lambda R(\lambda)x - x$, so $PHx = \frac{1}{2\pi i} \oint_{\Gamma} (\lambda R(\lambda)x - x) d\lambda = HPx$. $M = \ker(I - P)$ is closed, and for $x \in M$, $Hx = HPx = PHx \in M$.

(b) If $x \neq 0$, then μ is an eigenvalue (with j the largest index such that $(H - \mu)^j x \neq 0$, $(H - \mu)^j x$ is an eigenvector), so $\mu \notin \Gamma$. For $\lambda \in \rho(H)$ let $y = \sum_{j=0}^{k-1} (\lambda - \mu)^{-j-1} (H - \mu)^j x \in D(H)$. Writing $\lambda - H = (\lambda - \mu) - (H - \mu)$, the sum telescopes: $(\lambda - H)y = x - (\lambda - \mu)^{-k} (H - \mu)^k x = x$, so $R(\lambda)x = y$. Integrating over Γ , $Px = \sum_j \left[\frac{1}{2\pi i} \oint_{\Gamma} (\lambda - \mu)^{-j-1} d\lambda \right] (H - \mu)^j x$, which is x if $\mu \in \Omega$ and 0 if $\mu \notin \bar{\Omega}$.

(c) By Lemmas A.2(b) and A.3(c), $Q(\zeta)X \subset D(H)$ and, using $\lambda(\lambda - \zeta)^{-1} = 1 + \zeta(\lambda - \zeta)^{-1}$ and $\frac{1}{2\pi i} \oint_{\Gamma} (\lambda - \zeta)^{-1} d\lambda = 1$,

$$HQ(\zeta) = \frac{1}{2\pi i} \oint_{\Gamma} \frac{\lambda R(\lambda) - I}{\lambda - \zeta} d\lambda = P + \zeta Q(\zeta) - I,$$

that is, $(\zeta - H)Q(\zeta) = I - P$. For $x \in D(H)$, $R(\lambda)Hx = HR(\lambda)x$ gives $Q(\zeta)Hx = HQ(\zeta)x$ in the same way, so $Q(\zeta)(\zeta - H)x = (\zeta - H)Q(\zeta)x = (I - P)x$.

(d) Let $T = H|_M$, a linear map of the finite-dimensional space M . An eigenvalue μ of T is an eigenvalue of H , so $\mu \notin \Gamma$; if $\mu \notin \bar{\Omega}$, an eigenvector $x \in M$ would satisfy $x = Px = 0$ by (b). So $\text{spec } T \subset \Omega \cap \text{spec } H$. Conversely, let $\zeta \in \Omega \setminus \text{spec } T$ and $S = (\zeta - T)^{-1}P + Q(\zeta)(I - P) \in B(X)$. For $x \in X$, $(\zeta - H)Sx = Px + (I - P)^2x = x$ by (c). For $x \in D(H)$, $(I - P)x \in D(H)$, $(\zeta - T)^{-1}P(\zeta - H)x = (\zeta - T)^{-1}(\zeta - T)Px = Px$ by (a), and $Q(\zeta)(I - P)(\zeta - H)x = Q(\zeta)(\zeta - H)(I - P)x = (I - P)x$ by (a) and (c); so $S(\zeta - H)x = x$. Hence $\zeta \in \rho(H)$, and $\text{spec } H \cap \Omega = \text{spec } T$, a finite set of eigenvalues of H . For $\mu \in \Omega$, (b) gives $G_{\mu}(H) \subset M$, so $G_{\mu}(H) = \bigcup_k \ker(T - \mu)^k = \ker(T - \mu)^{\dim M}$, which is $\{0\}$ unless $\mu \in \text{spec } T$. The decomposition of M under T gives $M = \bigoplus_{\mu \in \text{spec } T} G_{\mu}(H)$, and $\text{rank } P = \dim M = n(H, \Omega)$.

(e) By Lemma A.3(b), P is approximable; let F have finite rank with $\|P - F\| < 1$. If $x \in M$ and $Fx = 0$, then $\|x\| = \|(P - F)x\| \leq \|P - F\| \|x\|$, so $x = 0$. Thus F is injective on M and $\dim M \leq \text{rank } F$. $\square$

Theorem A.5 (compact resolvent). *Let $X = \ell_w^1(\mathcal{J})$ and let H have compact resolvent. Then:*

- (i) $R(\lambda)$ is compact and approximable for every $\lambda \in \rho(H)$;
- (ii) $\text{spec } H$ consists of eigenvalues and has no accumulation point in $\mathbb{C}$;
- (iii) for every $\mu \in \mathbb{C}$, $m = m(\mu; H)$ is finite, H maps $G_{\mu}(H)$ into itself, and $G_{\mu}(H) = \ker(H - \mu)^m$, so that $(H - \mu)|_{G_{\mu}(H)}$ is nilpotent;
- (iv) for every rectangle Ω with $\Gamma \subset \rho(H)$, Ω contains finitely many eigenvalues, and the Riesz projection $\frac{1}{2\pi i} \oint_{\Gamma} R(\lambda) d\lambda$ is a projection onto $\bigoplus_{\mu \in \Omega} G_{\mu}(H)$ of rank $n(H, \Omega) < \infty$.

Proof. (i) Lemmas A.1 and A.2(d); approximable operators are compact, as norm limits of compact operators.

(ii) Fix $\lambda_0 \in \rho(H)$, $\mathcal{K} = R(\lambda_0)$ and $r > 0$; we show that $\text{spec } H \cap \{|\mu - \lambda_0| \leq r\}$ is a finite set of eigenvalues, which proves (ii) since r is arbitrary. By (i) there is a finite-rank F with $e := \|\mathcal{K} - F\| < 1/r$; let $\Delta = \mathcal{K} - F$. For $\mu \neq \lambda_0$ put $\zeta = (\lambda_0 - \mu)^{-1}$. Then, for $x \in D(H)$,

$$(\mu - H)x = (\lambda_0 - \mu)(\zeta - \mathcal{K})(\lambda_0 - H)x, \quad (14)$$

because $(\lambda_0 - \mu)\mathcal{K}(\lambda_0 - H)x = (\lambda_0 - \mu)x$. On $Z = \{\zeta : |\zeta| > e\}$, $\zeta - \Delta$ is invertible with $(\zeta - \Delta)^{-1} = \sum_{k \geq 0} \Delta^k \zeta^{-k-1}$, which is holomorphic and of norm at most $(|\zeta| - e)^{-1}$, and $\zeta - \mathcal{K} = (\zeta - \Delta)(I - G(\zeta))$ with $G(\zeta) = (\zeta - \Delta)^{-1}F$. Choose a basis $f_1, \dots, f_n$ of FX and write $Fx = \sum_j \varphi_j(x)f_j$; the φ_j are bounded linear functionals, because coordinates on a finite-dimensional space are bounded. Then $G(\zeta)x = \sum_j \varphi_j(x)g_j(\zeta)$ with $g_j(\zeta) = (\zeta - \Delta)^{-1}f_j$. Let $\Phi(\zeta)$ be the $n \times n$ matrix with entries $\varphi_i(g_j(\zeta))$ and $d(\zeta) = \det(I - \Phi(\zeta))$, holomorphic on Z ; as $|\Phi_{ij}(\zeta)| \leq \|\varphi_i\| \|f_j\| / (|\zeta| - e)$, $d(\zeta) \rightarrow 1$ as $|\zeta| \rightarrow \infty$.

If $d(\zeta) \neq 0$, then $I - G(\zeta)$ is invertible: a solution of $x - G(\zeta)x = y$ has the form $x = y + \sum_j c_j g_j(\zeta)$ with $c_i = \varphi_i(x)$, and applying φ_i shows $(I - \Phi(\zeta))c = (\varphi_i(y))_i$; conversely, with $c = (I - \Phi(\zeta))^{-1}(\varphi_i(y))_i$, the vector $x = y + \sum_j c_j g_j(\zeta)$ has $\varphi_i(x) = \varphi_i(y) + (\Phi(\zeta)c)_i = c_i$ and so solves it. So the solution exists, is unique, and depends boundedly on y . If $d(\zeta) = 0$, take $c \neq 0$ with $(I - \Phi(\zeta))c = 0$ and $x = \sum_j c_j g_j(\zeta)$: then $\varphi_i(x) = c_i$, so $x \neq 0$ and $x - G(\zeta)x = 0$.

Now let $0 < |\mu - \lambda_0| \leq r$, so that $|\zeta| \geq 1/r > e$ and $\zeta \in Z$. If $d(\zeta) \neq 0$, then $\zeta - \mathcal{K}$ is invertible and by (14) $\mu - H$ is a bijection $D(H) \rightarrow X$ with the bounded inverse $(\lambda_0 - \mu)^{-1}\mathcal{K}(\zeta - \mathcal{K})^{-1}$, so $\mu \in \rho(H)$. If $d(\zeta) = 0$, there is $y \neq 0$ with $(\zeta - \mathcal{K})y = (\zeta - \Delta)(I - G(\zeta))y = 0$; then $x = \mathcal{K}y = \zeta y \neq 0$ lies in $D(H)$, and by (14) $(\mu - H)x = (\lambda_0 - \mu)(\zeta - \mathcal{K})y = 0$, so μ is an eigenvalue. Finally, the set $\{\zeta : |\zeta| \geq 1/r, d(\zeta) = 0\}$ is closed, and bounded because $d \rightarrow 1$ at infinity, so it is a compact subset of the connected open set Z , on which d is holomorphic and not identically zero; by the identity theorem it is finite. Since $\mu \mapsto \zeta$ is a bijection of $\{0 < |\mu - \lambda_0| \leq r\}$ onto $\{|\zeta| \geq 1/r\}$ and $\lambda_0 \in \rho(H)$, the claim follows.

(iii) If μ is not an eigenvalue, $G_\mu(H) = \{0\}$ (a nonzero element of $\ker(H - \mu)^k$ produces an eigenvector, as in the proof of Proposition A.4(b)). Otherwise, by (ii), there is an open square Ω centred at μ with $\overline{\Omega} \cap \text{spec } H = \{\mu\}$, so $\Gamma \subset \rho(H)$. By (i) and Proposition A.4(a), (d) and (e), $M = PX = G_\mu(H)$ is finite-dimensional, $H(M) \subset M$, and the only eigenvalue of $H|_M$ is μ , so $(H|_M - \mu)^{\dim M} = 0$ and $G_\mu(H) = \ker(H - \mu)^m$.

(iv) Proposition A.4(d) and (e), with (i). □

Lemma A.6 (continuity of the Riesz projection). *Let H_0 be an operator in X , $\mathcal{E} \in B(X)$, $H_1 = H_0 + \mathcal{E}$, Ω a rectangle with $\Gamma \subset \rho(H_0) \cap \rho(H_1)$, and $\|(\lambda - H_k)^{-1}\| \leq C$ on Γ for $k = 0, 1$. Then the Riesz projections P_k of H_k for Ω satisfy $\|P_1 - P_0\| \leq \frac{|\Gamma|}{2\pi} C^2 \|\mathcal{E}\|$.*

Proof. $\rho(H_0) \neq \emptyset$, and by Lemma A.2(e) applied to H_0 and $\mathcal{E}$, $(\lambda - H_1)^{-1} - (\lambda - H_0)^{-1} = (\lambda - H_1)^{-1}\mathcal{E}(\lambda - H_0)^{-1}$ on Γ ; integrate (Lemma A.3(a)). □

Theorem A.7 (the count is invariant under a homotopy). *Let $X = \ell_w^1(\mathcal{J})$, $\hat{D}$ an operator with compact resolvent, $\hat{E} \in B(X)$, Ω a rectangle with $\Gamma \subset \rho(\hat{D})$, and suppose that $I - s\hat{E}(\mu - \hat{D})^{-1}$ is invertible for every $(s, \mu) \in [0, 1] \times \Gamma$. Let $H(s) = \hat{D} + s\hat{E}$. Then for every $s \in [0, 1]$, $\Gamma \subset \rho(H(s))$, $H(s)$ has compact resolvent, and $n(H(s), \Omega)$ is finite and does not depend on s .*

Proof. By Lemma A.2(e), every $\mu \in \Gamma$ lies in $\rho(H(s))$ and $R_s(\mu) := (\mu - H(s))^{-1} = R_{\hat{D}}(\mu)(I - s\hat{E}R_{\hat{D}}(\mu))^{-1}$, which is compact by Theorem A.5(i) for $\hat{D}$. The map $(s, \mu) \mapsto I - s\hat{E}R_{\hat{D}}(\mu)$ is

norm continuous on $[0, 1] \times \Gamma$ (Lemma A.2(c)) with invertible values, and inversion is continuous on the invertible operators (if $\|B - A\|\|A^{-1}\| < 1$, then B is invertible and $\|B^{-1} - A^{-1}\| \leq \|A^{-1}\|^2\|B - A\|/(1 - \|A^{-1}\|\|B - A\|)$, by the Neumann series). So $R_s(\mu)$ is continuous on the compact set $[0, 1] \times \Gamma$ and bounded there by some C . By Theorem A.5(iv), the Riesz projection $P(s)$ of $H(s)$ for Ω has rank $n(H(s), \Omega) < \infty$, and by Lemma A.6 with $H_0 = H(s)$ and $\mathcal{E} = (s' - s)\hat{E}$, $\|P(s) - P(s')\| \leq \frac{|\Gamma|}{2\pi} C^2 \|\hat{E}\| |s - s'|$. When this is less than 1, Lemma 5.8 gives $\text{rank } P(s) = \text{rank } P(s')$. So the rank is locally constant on the connected interval $[0, 1]$, hence constant. $\square$